

BiALIGN PYRAMID: A Design Space of Human–AI Alignment Mechanisms Across Scales

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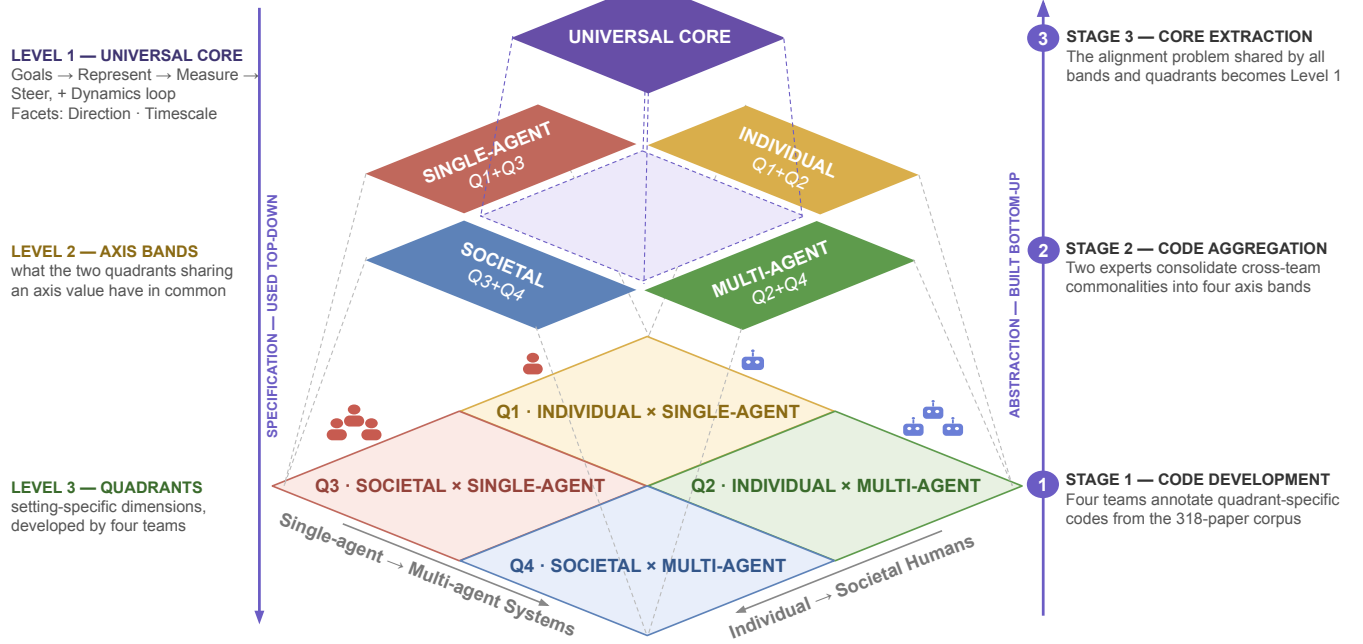


Figure 1: The BiALIGN PYRAMID, a three-level design space for human–AI alignment mechanisms organized along two axes: human scale (individual→societal) and AI configuration (single-agent→multi-agent). Level 1 provides a theory-informed Universal Core comprising the complementary functions of Represent, Measure, and Steer, together with the cross-cutting facets of *Direction* and *Timescale*. These functions describe how mechanisms contribute to a potential system-level alignment loop; individual mechanisms need not implement or connect all three. Level 2 contains four Axis Bands that capture concerns shared by quadrants with the same value on one axis. Level 3 contains four Quadrants (Q1–Q4) and their setting-specific mechanism dimensions. We developed the space by coding the 318-paper corpus within quadrants (Stage 1), synthesizing recurring concerns into Axis Bands (Stage 2), and confirming and refining the theory-informed core across settings (Stage 3). The pyramid is constructed bottom-up through synthesis and applied top-down to characterize and compare alignment mechanisms.

Abstract

Human–AI alignment research spans heterogeneous mechanisms across separate communities and settings, complicating comparisons of their assumptions, functions, and applicability. We introduce the BiALIGN PYRAMID, a multilevel design space organized along two dimensions: human scale, from individual to societal, and AI configuration, from single-agent to multi-agent systems. From

89,241 papers across eleven venues, 26 interdisciplinary researchers constructed and qualitatively coded a corpus of 318 papers. The design space comprises three levels: 1) a Universal Core comprising the Represent–Measure–Steer functional loop, characterized by Direction and Timescale; 2) four Axis Bands capturing concerns shared across settings; and 3) four Quadrants containing configuration-specific mechanisms. Mapping the analyzed corpus reveals differences in how alignment is operationalized and limited coverage of reciprocal adaptation, longitudinal feedback, and representations that affected people can inspect and revise. The BiALIGN PYRAMID provides a shared vocabulary for comparing mechanisms, clarifying design choices, and identifying important opportunities for future human–AI interaction research.

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[†]Superscripts denote author roles: 1 for four teams’ leaders, 2 for team members, 3 for project advisors, and 4 for the project leader & principal investigator. Within each role, authors contributed equally and are listed alphabetically by last name. Please see the full author list with roles, affiliations, and contributions in Appendix A.

CCS Concepts

• **Human-centered computing** → **Interaction design theory, concepts and paradigms**; **HCI theory, concepts and models**; **Collaborative and social computing theory, concepts and paradigms**.

1 Introduction

Human–AI alignment is pursued through a growing and heterogeneous range of mechanisms. In domains such as education, healthcare, and scientific discovery, AI systems increasingly participate in consequential interactions with people [71, 120, 206]. Alignment research includes mechanisms for eliciting human goals, adapting model objectives, evaluating system behavior and outcomes, supporting appropriate reliance, and enabling human or institutional intervention. Human feedback may reshape AI objectives and policies, while interaction with AI may influence people’s behavior, beliefs, preferences, and reliance [21]. These mechanisms are studied across communities that emphasize different parts of the problem: ML examines preference optimization, constitutional approaches, and policy updates [9, 21, 176]; NLP investigates feedback learning, preference data, and pluralistic alignment [106, 187, 211]; HCI studies interfaces, trust calibration, reliance, and people’s understanding of AI systems [3, 17]; and responsible AI contributes participatory elicitation, collective input, auditing, and governance [81, 240]. Because these communities use different vocabularies and focus on different human and AI configurations, researchers lack a shared structure for comparing mechanisms, examining their assumptions, and determining where an approach may transfer across settings.

To connect this fragmented literature, we ask: **How do existing human–AI alignment mechanisms represent alignment goals, measure alignment, and steer human–AI systems across different human scales and AI configurations?** We address this question through developing a multilevel design space for human–AI alignment mechanisms. A design space identifies the dimensions along which a class of designs can vary [19, 20, 117]. Here, it serves three purposes: (i) providing a common vocabulary for comparing related mechanisms across research communities; (ii) making explicit whose states a mechanism represents, what it measures, which part of the human–AI system it steers, and over what timescale; and (iii) revealing combinations of functions, configurations, and interventions that receive limited coverage in the analyzed corpus. The space thereby helps explain why a mechanism may suit one setting but not transfer directly to another, while indicating what mechanisms for underexplored settings may need to address. We treat direction, timescale, and loop closure as properties that vary across mechanisms rather than assuming that every alignment mechanism is bidirectional, longitudinal closed.

The resulting framework, BiALIGN PYRAMID (Fig. 1), is a three-level design space organized along two axes: human scale (individual → societal) and AI configuration (single-agent → multi-agent systems). Crossing these axes produces four configurations: **individual × single-agent**, **individual × multi-agent**, **societal × single-agent**, and **societal × multi-agent**. At Level 1, a theory-informed **Universal Core** comprises three complementary functions: REPRESENT, MEASURE, and STEER [197]. These functions characterize how mechanisms capture alignment-relevant goals or states, assess

behavior or outcomes, and influence subsequent system behavior. Together they describe a potential system-level functional loop, although an individual mechanism may implement one or more functions without connecting all three. Level 2 contains four **Axis Bands**—Individual, Societal, Single-Agent, and Multi-Agent—that capture concerns shared by configurations with the same value on one axis. Level 3 contains four **Quadrants** and their configuration-specific mechanism dimensions. Two cross-cutting facets further characterize each mechanism: *Direction* indicates whether it primarily targets the AI, the human, or connects changes on both sides through mutual adaptation, while *Timescale* indicates whether it operates at the turn, session, or longitudinal level.

We developed the BiALIGN PYRAMID through a theory-informed and empirically elaborated process. Following a PRISMA-guided review, we retrieved 89,241 papers from eleven venues spanning HCI, NLP, ML, and responsible AI. Keyword screening, conceptual eligibility assessment, centrality classification, and coverage-preserving sampling produced a focused corpus of 318 papers. 26 interdisciplinary researchers, organized into four teams, qualitatively coded this corpus. The teams developed Level 3 mechanism dimensions within each quadrant; recurring concerns were synthesized into the Level 2 Axis Bands; and the theory-informed Represent–Measure–Steer scaffold [197] was confirmed and refined across configurations. Stage-balanced mean Cohen’s κ for human validation ranged from .752 to .926 across quadrants. Mapping the analyzed corpus shows that alignment is operationalized differently across human scales and AI configurations. It also identifies comparatively limited coverage of mechanisms that connect changes on both sides of the human–AI system, sustain operational feedback across repeated interactions, or provide representations that affected people can inspect and revise. We interpret these results as patterns within the analyzed corpus rather than unbiased estimates of prevalence across all alignment research.

This work makes three contributions. First, we contribute a **multilevel design space of human–AI alignment mechanisms**, comprising the complementary functions of Represent, Measure, and Steer, four cross-setting Axis Bands, four configuration-specific Quadrants, and the cross-cutting facets of Direction and Timescale. Second, we provide a **systematic map across human and AI configurations**, showing how mechanisms vary from individual to societal settings and from single- to multi-agent AI systems. Third, we provide an **empirical analysis of coverage and design opportunities**, identifying underexplored mechanism combinations and translating them into opportunities for reciprocal adaptation, inspectable and revisable representations, participatory governance, and feedback across repeated interactions. The BiALIGN PYRAMID is neither exhaustive nor prescriptive; rather, it provides researchers, designers, and policymakers with a shared vocabulary for comparing mechanisms and identifying what may be missing from a human–AI system. We release the annotated corpus and codebook as a living artifact, enabling the community to audit, refine, and extend the design space.

2 Background

This section establishes the analytical dimensions of the BiALIGN PYRAMID. We first distinguish the functions alignment mechanisms

Table 1: Positioning the BiALIGN PYRAMID relative to prior alignment frameworks.

Framework	Functional scaffold	Human scale	AI configuration	Corpus mapping
Shen et al. [197]	✓	Partial	Partial	–
Human–agent alignment [62]	Partial	Individual	Single-agent	–
Prior technique/risk surveys	Varies	Varies	Varies	Partial
BiAlign Pyramid	✓	Individual–societal	Single–multi	✓

perform, then separate direction from temporal horizon and feedback. We next motivate human scale and AI configuration as the two organizing axes, before positioning our contribution relative to existing frameworks.

2.1 Functions of Alignment Mechanisms

Human–AI alignment is pursued through mechanisms that represent alignment targets, assess whether alignment holds, or intervene in subsequent behavior. Technical approaches such as reinforcement learning from human feedback, direct preference optimization, and constitutional AI translate human-provided signals into changes in model behavior [9, 155, 176]. HCI and responsible AI further contribute mechanisms for elicitation, evaluation, interactive correction, user control, participatory input, and institutional oversight [3, 62, 81, 240]. Building on prior functional accounts of alignment [197], we use **REPRESENT**, **MEASURE**, and **STEER** as a theory-informed scaffold. **REPRESENT** captures alignment-relevant goals, values, states, relationships, or contexts; **MEASURE** assesses behavior, outcomes, experiences, or possible misalignment; and **STEER** intervenes in the AI, the interaction, or the surrounding institution. These are complementary functions that every mechanism must implement. At the system level they may be connected through feedback, but an individual mechanism may contribute to only one or two functions.

2.2 Direction, Timescale, and Feedback

Alignment mechanisms also differ in whom they are designed to change. Preference learning, interactive correction, and personalization primarily translate human input into changes in AI behavior [21, 155, 176]. Explanations, decision support, and calibration mechanisms may instead affect people’s understanding, reliance, or behavior [3, 17]. Other mechanisms explicitly connect changes on both sides through mutual adaptation [197, 198]. We characterize these alternatives through **Direction**: Human→AI, AI→Human, or Mutual. Direction is distinct from **Timescale**, which records whether a mechanism explicitly models change at the turn, session, or longitudinal level. A longitudinal mechanism may repeatedly update only the AI, while influence in both directions may occur without being connected through sustained feedback. Likewise, observing change over time does not mean that the observation informs subsequent representation, measurement, or steering. Direction, Timescale, and loop closure therefore capture related but distinct properties. They do not imply that every mechanism operates in both directions or closes a longitudinal feedback loop.

2.3 Alignment Across Human and AI Configurations

Alignment changes as the relevant human and AI configurations expand. At the **individual** level, mechanisms commonly address personal goals, understanding, reliance, and control. At the **societal** level, they must account for plural and contested values, unequal effects, collective input, and legitimate authority [81, 187, 211]. Similarly, a **single AI system** provides a relatively centralized locus of behavior and intervention, whereas **multi-agent systems** introduce role allocation, orchestration, interaction effects, and distributed responsibility. Human scale and AI configuration define the setting in which a mechanism operates and therefore determine the actors, decision structures, and governance relationships involved. Direction and Timescale instead characterize properties that can vary within every setting. We therefore use human scale and AI configuration as organizing axes and Direction and Timescale as cross-cutting facets. Crossing the axes produces four configurations: individual × single-agent, individual × multi-agent, societal × single-agent, and societal × multi-agent.

2.4 Relationship to Existing Frameworks

Prior work provides important but partial views of this landscape (Table 1). Shen et al. [197] articulate functional and directional aspects of human–AI alignment, providing the theoretical starting point for our Represent–Measure–Steer scaffold. Work on human–agent alignment examines what people and AI systems must coordinate during interaction [62], while other surveys organize alignment around particular techniques, values, risks, or application settings. These accounts clarify individual components of alignment but **do not systematically compare mechanisms across both human scale and AI configuration**. The BiALIGN PYRAMID extends this work in three ways. First, it maps mechanisms across the individual–societal and single–multi-agent axes. Second, it distinguishes shared Axis Bands from configuration-specific Quadrant dimensions. Third, it grounds this structure in a coded corpus and uses Direction and Timescale to compare coverage across settings. Represent, Measure, and Steer are therefore theory-informed; the Axis Bands, Quadrant codebooks, and corpus-level mapping constitute the principal empirical extensions.

3 Approach

We developed the BiALIGN PYRAMID through a theory-informed and empirically elaborated review process. We first defined the unit and scope of analysis, constructed a focused corpus through a PRISMA-guided review, developed configuration-specific mechanism dimensions through qualitative coding, and synthesized these

PRISMA-guided corpus construction for qualitative coding

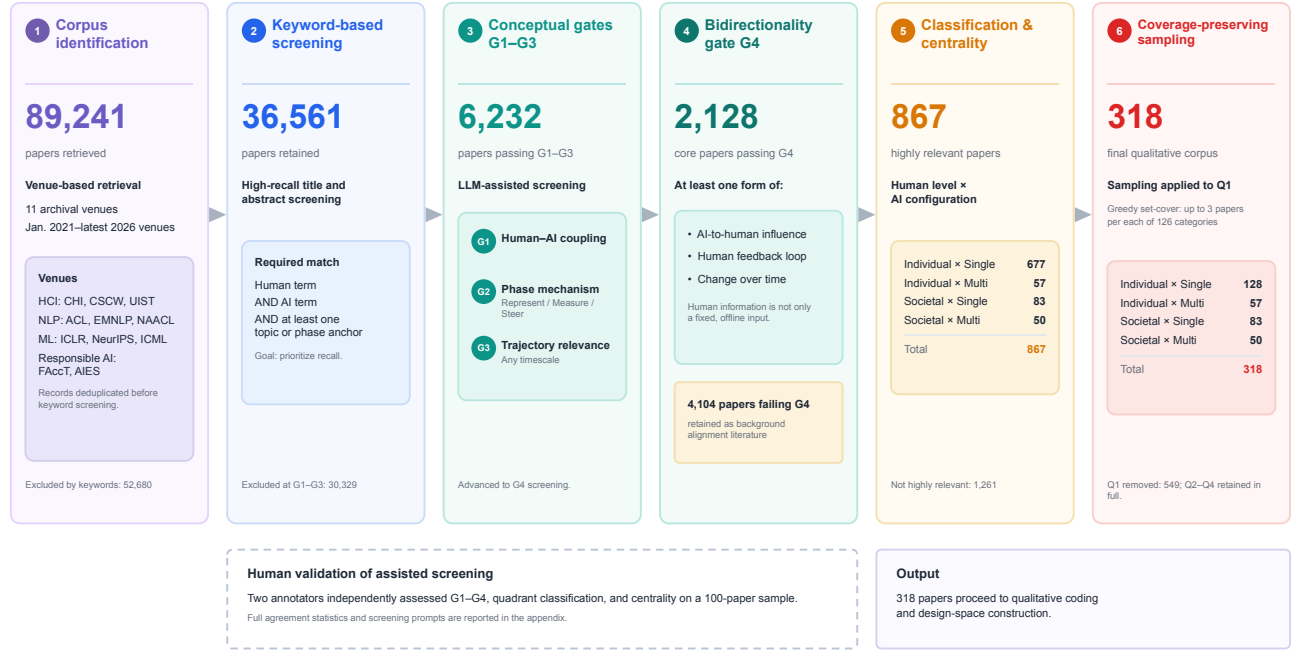


Figure 2: Overview of the paper identification, screening, centrality assessment, and coverage-preserving sampling process used to construct the final corpus.

dimensions into the three levels of the pyramid. Figure 2 summarizes the corpus-construction pipeline.

3.1 Scope and Unit of Analysis

Alignment mechanisms. We define *human–AI alignment* as the extent to which the goals, behavior, and outcomes of a human–AI system remain consistent with human values and agency at the individual or collective level. Alignment may be pursued through mechanisms that operate on the AI, the human, their relationship, or the surrounding social and institutional context. This definition does not require every mechanism to operate bidirectionally, longitudinally, or through a closed feedback loop.

Our unit of analysis is the *mechanism*: a concrete and functionally identifiable component, method, procedure, or intervention that contributes to representing alignment-relevant states, assessing whether alignment holds, or influencing subsequent system behavior. We coded a mechanism for REPRESENT if it captured an alignment-relevant state or target, MEASURE if it assessed alignment-relevant evidence, and STEER if it intervened in subsequent behavior or organization. A paper could contribute multiple mechanisms, and a mechanism could implement one or more functions. The design space classifies mechanisms; paper-level labels are used separately for corpus-level summaries of Direction and Timescale.

Two axes and four quadrants. We situated each mechanism along two axes: human scale (individual or societal) and AI configuration (single- or multi-agent). Crossing the axes yields four

quadrants: Q1, an individual and a focal AI system; Q2, an individual and multiple interacting agents; Q3, a collective and a focal AI system; and Q4, collectives and multi-agent systems. Mechanisms were assigned according to the configuration for which they were designed, rather than every setting to which their effects might propagate. Real systems may span or transition between these analytical categories, a limitation we revisit in Section 5.5.

Scope boundaries. A paper was in scope if it: (1) involved humans and AI or used a human-derived normative standard as an alignment target; (2) contributed a concrete technique, system, method, framework, or measurement instrument rather than observation alone; (3) addressed an alignment-relevant relationship at an identifiable temporal horizon; and (4) involved interaction, AI-to-human influence, feedback from human responses, or temporal change beyond treating human information solely as a fixed offline input. These criteria produced a focused corpus of interactive or adaptive alignment mechanisms, not a representative sample of all alignment research.

3.2 PRISMA-Guided Systematic Review

Identification and screening. Following the PRISMA guidelines [158, 213] and established recommendations for literature reviews in HCI [213], we retrieved 89,241 papers published from January 2021 through the latest available 2026 proceedings of eleven venues spanning HCI (CHI, CSCW, UIST), NLP (ACL, EMNLP, NAACL), machine learning (ICLR, NeurIPS, ICML), and responsible AI (FAccT,

AIES); Figure 2 summarizes the pipeline, and Appendix B.2 details venues and retrieval sources. A high-recall keyword screen over titles and abstracts — requiring a human term, an AI term, and at least one topic or stage anchor — retained 36,561 papers (keyword set: Appendix B.2–B.4).

Conceptual gates. We then applied the four scope criteria of Section 3.1 as screening gates: *G1* human–AI coupling, *G2* a Represent/Measure/Steer mechanism contribution, *G3* trajectory relevance at any timescale, and *G4* interaction and adaptation relevance (definitions: Appendix B.4). Gates *G1–G3* yielded 6,232 papers, and *G4* narrowed these to a core corpus of 2,128; papers failing only *G4* were retained as background alignment literature. Gate screening was LLM-assisted at scale and validated by two human annotators on a 100-paper sample: human–human agreement across gates was 0.80–0.88 (Gwet’s AC1 0.77–0.87) and human–AI agreement 0.77–0.96 (AC1 0.74–0.96), supporting the assisted protocol (prompts, procedure, and full results: Appendix C).

Classification and sampling. Each core paper was classified along the two axes of Section 3.1 and rated for centrality to bidirectional and dynamic alignment, identifying 867 highly relevant papers: 677 individual×single-agent, 57 individual×multi-agent, 83 societal×single-agent, and 50 societal×multi-agent. Because the first group dwarfed the others, we applied coverage-preserving sampling to it alone: its 677 papers were organized into 126 fine-grained categories, and a greedy set-cover selected up to three representatives per category (128 papers), while every paper in the other three quadrants was retained. The final corpus comprises 318 papers for qualitative coding.

3.3 Constructing the Design Space

We combined a theory-informed functional scaffold with qualitative code development. Represent, Measure, and Steer were motivated by prior work [197] and provided an initial organization for coding. Level 3 mechanism dimensions were developed within quadrants, Level 2 Axis Bands were synthesized across related quadrants, and Level 1 was confirmed and refined against the finalized mechanism codes.

Stage 1: Quadrant dimensions (Level 3). We organized qualitative coding into four interdisciplinary teams corresponding to the four quadrants. The 26 annotators included graduate students, postdoctoral researchers, and faculty with expertise in HCI, NLP, cognitive science, responsible AI, and machine learning. Each team had a lead annotator who coordinated calibration, maintained its codebook, and resolved procedural questions. Before primary annotation, annotators received shared coding guidelines and completed a calibration round of 5–20 papers. They then reviewed relevant paper sections, identified alignment mechanisms, and generated codes for their assigned functions without viewing AI-generated candidate labels. After this independent pass, annotators compared their codes with the candidate labels, adding relevant concepts they had missed, merging semantically equivalent labels, or rejecting unsupported suggestions. The teams subsequently consolidated overlapping concepts and harmonized their definitions to form the Level 3 codebooks. Using the finalized codebooks, papers were recoded for their assigned functions. Q1 and Q2 papers were assessed across Represent, Measure, and Steer, whereas Q3 and Q4

papers were allocated to a primary function during the original coding process. We therefore calculate each distribution using its corresponding coding denominator and do not interpret differences in stage-level frequencies across these groups as direct prevalence comparisons.

Stage 2: Axis Bands (Level 2). Two senior researchers compared the finalized Level 3 codebooks across quadrants. They harmonized differences in terminology and granularity and identified mechanism families shared by the two quadrants agreeing on one axis value. A theme was assigned to an Axis Band when it captured the same underlying alignment concern in both relevant quadrants, even when its implementation differed across configurations. This procedure produced four bands: Individual, Societal, Single-Agent, and Multi-Agent. Candidate bands and their supporting Level 3 codes were reviewed with the relevant quadrant teams. Disagreements concerned whether similar labels represented a shared underlying concern or only a superficial terminological overlap. The researchers resolved these cases by comparing code definitions, supporting passages, and representative mechanisms. The released codebook records the Level 3 dimensions supporting each Axis Band.

Stage 3: Universal Core confirmation and refinement (Level 1). Finally, we assessed whether every finalized Level 3 code could be assigned to at least one of the theory-informed Represent, Measure, or Steer functions [197]. We traced each code to its function, reviewed ambiguous placements, and refined the corresponding decision rules. All finalized mechanism codes could be characterized by at least one function. This procedure confirmed the scaffold’s coverage without implying that every mechanism implements all three functions or forms a closed loop. We also applied two cross-cutting facets. Direction was coded as Human→AI when the designed end-point of change was the AI, AI→Human when it was the human, and Mutual only when changes on both sides were explicitly connected by the mechanism. Timescale was coded as turn, session, longitudinal, or unclear according to the longest temporal horizon explicitly modeled. These facets characterize mechanisms rather than serving as inclusion requirements.

The resulting three-level structure follows a scope-based placement rule. A function or facet applicable across all configurations belongs to Level 1; a concern shared by the two quadrants agreeing on one axis belongs to a Level 2 Axis Band; and a configuration-specific mechanism dimension belongs to a Level 3 Quadrant. Thus, Level 1 is theory-informed and corpus-confirmed, Levels 2 and 3 are derived through cross-quadrant and within-quadrant qualitative synthesis.

3.4 Reliability and Validation

We evaluated screening, cross-cutting facets, and Level 3 mechanism coding separately because they use different units and decision rules.

Screening validation. Two researchers independently assessed *G1–G4*, quadrant classification, and centrality for a 100-paper sample. Because the gate and centrality labels were imbalanced, we report observed agreement and Gwet’s AC1 for these assessments; for quadrant classification, we report observed agreement and Cohen’s κ . Human–human agreement across *G1–G4* ranged from .80

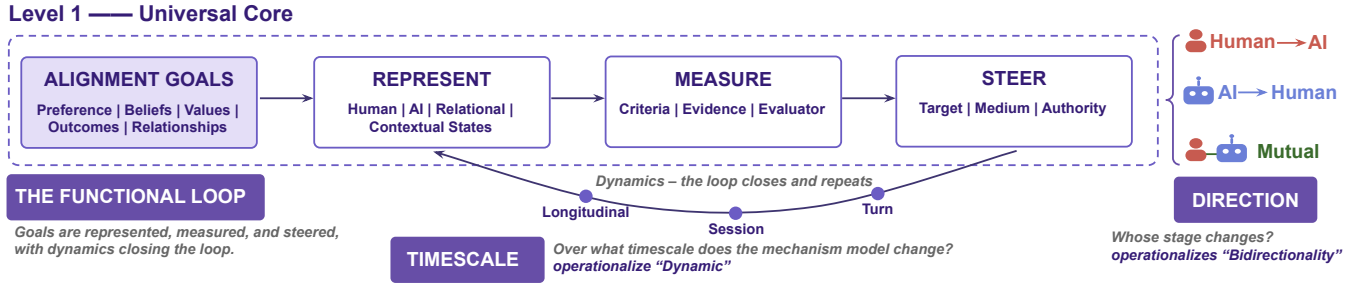


Figure 3: Level 1 of the BiALIGN PYRAMID: the Universal Core. REPRESENT, MEASURE, and STEER are complementary functions through which mechanisms capture alignment-relevant goals and states, assess relevant evidence, and influence subsequent behavior. At the system level, these functions may connect and repeat, but an individual mechanism need not implement all three or close the loop. Two cross-cutting facets further characterize each mechanism: *Direction* identifies its primary endpoint of influence (Human→AI, AI→Human, or Mutual), and *Timescale* identifies its explicit temporal horizon (turn, session, or longitudinal).

to .88, with AC1 ranging from .766 to .871. Human–AI agreement ranged from .77 to .96, with AC1 ranging from .743 to .959. Full results appear in Appendix B.1.5.

The validation sample was drawn from papers retained by the assisted screening process. These results therefore assess classification consistency among retained papers but do not estimate the false-negative rate among excluded papers. We account for this limitation when interpreting the corpus and avoid treating it as exhaustive.

Direction and Timescale reliability. Direction and Timescale labels were seeded from the screening outputs and reviewed using the finalized decision rules. To validate these facets, two annotators independently coded a stratified sample of 100 papers covering all four quadrants. Cohen’s κ was .78 for Direction and .72 for Timescale. Disagreements were resolved through discussion using the paper evidence and decision rules.

Level 3 coding reliability. Papers in the reliability sample were independently coded by a second annotator using the finalized Level 3 codebooks. Because coding was multi-label, we treated each code as a binary present/absent decision, calculated Cohen’s κ for every non-constant code, and macro-averaged the coefficients within each function and quadrant. The stage-balanced mean κ was .752 for Q1, .926 for Q2, .853 for Q3, and .866 for Q4. More details are reported in Appendix B.1.5.

Open materials. We will release the annotated corpus, codebooks, screening criteria, prompts, coding questions, and decision rules as a living artifact. These materials allow readers to audit placements, examine the relationship among the three levels, and apply or extend the design space as new alignment mechanisms emerge.

4 The BiALIGN PYRAMID Design Space

The design space organizes the 318-paper corpus into three levels of scope (Fig. 1). The Universal Core (Level 1, Section 4.1) provides an analytical scaffold applicable across all four quadrants; each Axis Band (Level 2, Section 4.2 captures concerns shared by two quadrants; and each Quadrant (Level 3, Section 4.3) contains setting-specific dimensions. The space was developed through

theory-informed, bottom-up synthesis (Section 3.3) and is applied top-down to characterize alignment mechanisms (Section 5.1).

4.1 Level 1: Universal Core

The Universal Core (Fig. 3) comprises three complementary functions—REPRESENT, MEASURE, and STEER—and two cross-cutting facets, *Direction* and *Timescale*.

A Functional Loop. The Universal Core (Fig. 3) comprises a functional loop in which goals are represented, measured, and steered, plus two cross-cutting facets: Direction and Timescale.

- **GOALS: what is aligned.** Four kinds of alignment targets recur across the corpus: *preferences and intentions*; *beliefs and understandings*; *values, social norms, and obligations*; and *outcomes and relationships*.
- **REPRESENT: whose state, in what form.** Representations encode four types of state: *human states* (preferences, intentions, behaviors, mental models), *AI states* (objectives, policies, internal representations, capabilities), *relational states* (mutual understanding, trust and reliance, roles and dependence, coordination and conflict), and *contextual states* (tasks, workflows, interaction history).
- **MEASURE: criterion, evidence, evaluator.** Every measurement mechanism answers three questions. Against what *criterion*—goal or preference satisfaction, relational or coordination quality, safety and harm, or change across time? From what *evidence*—self-report or deliberated judgment, human behavior and interaction traces, AI behavior or outputs, model-internal signals, or objective task outcomes? And judged by which *evaluator*—humans, or AI and LLM judges?
- **STEER: target, medium, authority.** Every steering mechanism likewise answers three questions. What *target* (AI state or behavior, human state or behavior, the human-AI relationship, or the interaction protocol)? Through what *medium* (prompting, context, or memory; training or model updating; information, explanation, or feedback; interface and interaction design; orchestration and role allocation; or incentives, rules, and governance)? Under whose *authority* (the end user, designer or developer, the AI system itself, or an organization or institution)?

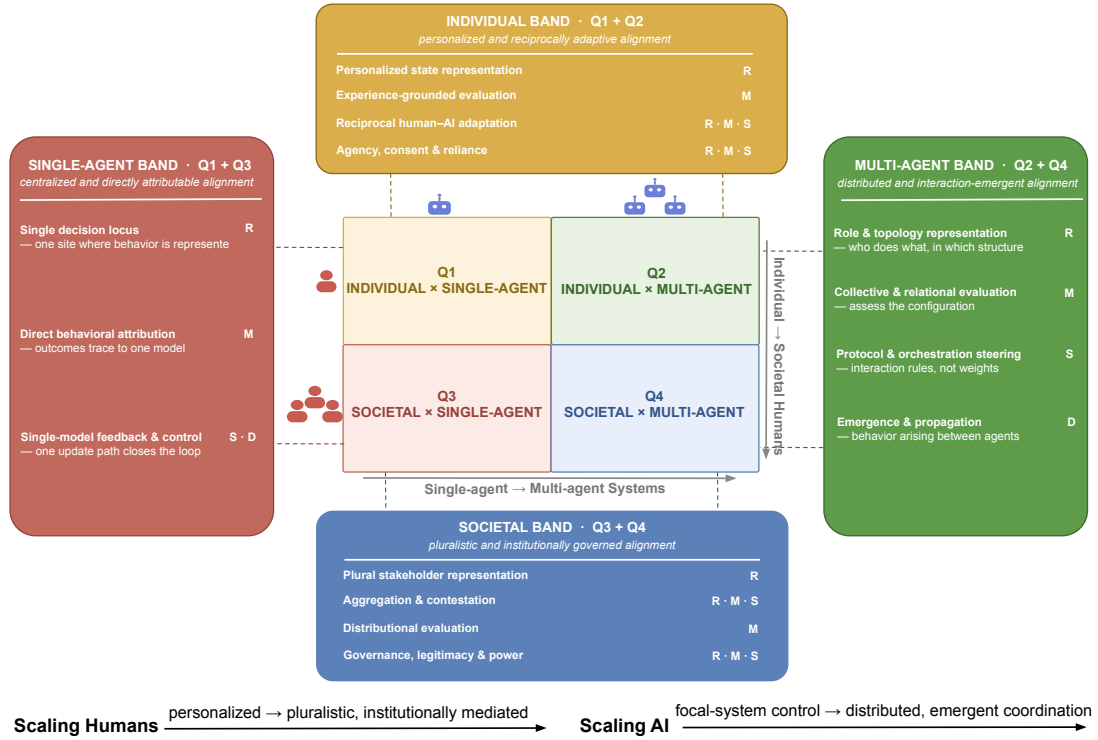


Figure 4: Level 2 of the BiALIGN PYRAMID: the four Axis Bands. Fixing one axis value fixes a family of concerns — each band holds in the two quadrants sharing that value, and every quadrant inherits one band per axis. Band dimensions are tagged by the Level-1 stage they refine (R/M/S/D). Read along either axis, the bands state the space’s central claim: scaling humans makes alignment pluralistic and institutionally mediated; scaling AI makes it distributed and interaction-emergent.

Facet 1: Direction. *Whose state is the mechanism designed to change?* Direction is a property of the whole mechanism for representing, measuring, and steering alignment. *Human→AI* mechanisms capture human states to change the AI’s behavior or representations. *AI→Human* mechanisms change the human side to support trust calibration and agency while mitigating overreliance. Finally, *Mutual* mechanisms close a single co-adaptive loop in which both sides update on each other.

Facet 2: Timescale. *Over what horizon does the mechanism explicitly model change?* A single *turn*, a *session*, or a *longitudinal* process spanning repeated use. This includes collective change such as norm shift, which is coded as longitudinal at the societal level.

4.2 Level 2: Four Axis Bands

Each band describes what alignment becomes once one axis value is fixed (individual, societal, single agent, multi-agent).

Individual Band (Q1+Q2): personalized and reciprocally adaptive alignment. When one person is the human side, mechanisms converge on *personalized state representation* [R] by modeling that person’s preferences, knowledge, and context. They also rely on *experience- and behavior-grounded evaluation* [M] measures against what the person does and reports rather than static benchmarks, as well as *reciprocal human–AI adaptation* [R/M/S] for mutual adjustment over use. Finally, these mechanisms leverage *agency, consent,*

and reliance [R/M/S] to keep a person’s control and trust as explicit design considerations.

Societal Band (Q3+Q4): pluralistic and institutionally governed alignment. With many individuals on the human side, mechanisms converge on *plural stakeholder representation* [R], encoding many parties’ values and their conflicts. They also rely on *aggregation and contestation* [R/M/S] by combining plural inputs while preserving channels to dispute the result, as well as *distributional evaluation* [M], measuring who is affected and how. Finally, they consider *governance, legitimacy, and power* [R/M/S], making explicit who may steer whom, on what authority.

Single-agent Band (Q1+Q3): centralized and directly attributable alignment. When one focal model is on the AI side, mechanisms share a *single decision locus* [R], where the model’s behavior is represented and attributed, as well as *direct behavioral attribution* [M], which traces the outcomes to that model’s outputs. Lastly, these mechanisms rely on *single-model feedback and control* [S/D], closing the loop through one update path.

Multi-agent Band (Q2+Q4): distributed and interaction-emergent alignment. When multiple agents interact, mechanisms share *role and topology representation* [R]. These mechanisms encode which agent does what within which structure. They also rely on *collective and relational evaluation* [M], assessing the configuration rather than any single agent. Finally, these mechanisms

	Code	Definition	References
REPRESENT	<i>Whose state is represented, and in what form?</i>		
	Human state representation	Representing alignment-relevant information about people: preferences, intentions, values, beliefs, knowledge, emotions, identity, behavior, capabilities, or interaction history. It can be elicited, inferred, or simulated.	[2, 126, 194]
	Context and memory construction	Using or constructing user- or task-specific histories, profiles, policies, or interaction context across time.	[55, 126, 149]
	Model-internal value representation	Representing alignment-relevant values, preferences, or objectives within the model’s learned latent or internal state space (e.g. AI training data, embedding, etc.).	[30, 121, 143]
	Co-maintained human-AI shared state	Maintaining a shared or co-created representation of goals, constraints, and interaction states accessible to both humans and AI systems.	[63, 82, 223]
MEASURE	<i>Against what criterion, on what evidence, judged by whom?</i>		
	User experience and self-report evaluation	Measuring human-reported experiences, perceptions, and attitudes toward AI interaction, including satisfaction, trust, understanding, perceived control, or well-being.	[10, 16, 180]
	Human behavioral evaluation	Inferring alignment from observable human behaviors, including user decisions, reliance, correction, or continued use.	[37, 73]
	Model state and response evaluation	Evaluating AI alignment through analysis of model states, internal representations, generated responses, or behavioral traces.	[50, 52, 95]
	Simulation-based judgement	Evaluating AI outputs or interactions using simulated users, stakeholder perspectives, or model-based evaluators such as LLM judges.	[22, 65]
	Human-AI calibration assessment	Measuring whether human and AI states, capabilities, expectations, or patterns of reliance are appropriately matched to support effective interaction.	[36, 149, 217]
STEER	Task performance evaluation	Evaluating alignment through objective task outcomes, using formula-based metrics (accuracy, success rate, error, reward) or estimation-based metrics, where task success serves as evidence of alignment between AI behavior and human goals.	[16, 90]
	<i>What is intervened on, through what channel, on whose authority?</i>		
	Prompt- and context-based adaptation	Adjusting AI behavior by modifying prompts, runtime context, contextual information, or interaction context provided to the focal model. (AI initiates steering based on context/prompt ...)	[40, 127, 248]
	Training or fine-tuning the focal model	Changing the parameters or behavior of the focal model through training procedures, fine-tuning, or data-based interventions. (requires access to model internals, permanent changes)	[94, 235, 249]
	Design recommendations for future aligned AI systems	Proposed: design recommendations, requirements or specifications that steer alignment of future versions of the current AI system design, or of future applications. Distinguishing criterion vs. Interface-based AI-behavior control: this code is for proposed/unimplemented specifications; if the paper implements a user-facing control, code it as Interface-based AI-behavior control instead.	[43, 133]
	Steering through overseer intervention	Steering through mechanisms for humans to intervene, override, or redirect AI agent decisions.	[70, 110, 123]
	Explanations, recommendations, and warnings	Providing explanations, recommendations, warnings, indicators or feedback that influence human understanding, decisions, or reliance on AI systems.	[26, 113, 212]
	Scaffolding human decisions or skills	Supporting human decision-making, learning, reflection, or skill development through AI assistance.	[47, 72, 185]
	Interface-based AI-behavior control	Providing interfaces that allow users to directly constrain, modify, or override AI outputs, actions, or operational behaviors (user initiates steering).	[253, 267, 284]

Table 2: Level-3 codes of Q1 (Individual $\times$ Single-Agent, 128 papers): mechanisms, codes, and definitions.

leverage *protocol- and orchestration-based steering* [S], intervening on interaction rules rather than model weights, and *emergence and propagation* [D], which are behavior that arises between agents.

4.3 Level 3: Four Quadrants

4.3.1 Q1 (Individual $\times$ Single-Agent): personalized focal-system alignment. Q1 covers the largest share of the corpus. We report the codes for Q1 in Table 2 and describe them in more detail below.

Represent. **How do mechanisms represent fine-grained, often specialized and evolving targets to align one individual and focal AI system?** The Represent dimension shows that the alignment aspects are rooted in either the system’s (mis) understanding of the human, or the human’s (mis)understanding the AI system. The Represent dimension contains four codes. *Human State Representation* models the person directly, capturing preferences, beliefs, knowledge, or behavior that are elicited from the person [87, 163, 209, 250], inferred from interaction traces or existing datasets [22, 103, 134, 241], or simulated [125, 252]. Meanwhile, *Context and memory construction* is agnostic about the person: it builds the alignment goal from the interaction context and may

keep a memory that evolves over time [70, 88, 102]. *Model-internal value representation* embeds the goal in the model’s own parameters, rewards, or activations, so that alignment depends on training data and internal representations rather than on externally supplied alignment data [1, 129, 143, 230]. Only *Co-maintained human-AI shared state* represents alignment in an artifact that the person and the system create and maintain together [36, 51, 63, 223].

Measure. **Which mechanisms can be used to measure (mis) alignment in both the individual and one focal AI system?** The Measure dimension in Q1 contains six codes. System-level mechanisms evaluate the single agent directly by inspecting its outputs and internals through *Model state and response evaluation* [33, 88, 95, 129], letting simulated users and LLM judges estimate alignment through *Simulation-based judgment* [65, 235, 264, 280], or using numerical *Task performance evaluation* metrics to evaluate alignment [16, 70, 178, 223]. Meanwhile, individual-level measures evaluate the person’s response, through *User experience and self-report evaluation* [10, 107, 115, 267, 269] and *Human behavioral evaluation* [16, 31, 47, 178]. Lastly, *Human-AI calibration assessment* measurement mechanisms evaluate whether a person’s trust, confidence, expectations, or reliance on the AI appropriately

matches the AI’s actual capability, accuracy, uncertainty, and limitations [14, 63, 113, 217].

Steer. How do mechanisms correct, adapt or influence an individual or focal AI system? The Steer dimension contained seven codes. Steering in Q1 acts overwhelmingly on the system, such as permanent and resource-intensive changes through *Training or fine-tuning the focal model* [13, 115, 250, 255], and non-permanent, execution-time changes through *Prompt- and context-based adaptation*, which modifies the prompt, the retrieved context or memory, or the model’s activations and decoding without touching its parameters [40, 80, 109, 134]. It also includes *Steering through overseer intervention*, where a human or an oversight system approves, corrects, or overrides the model’s actions during operation [31, 51, 70, 217], as well as *Design recommendations for future aligned AI systems* which direct steering at future systems or future iterations of the current one rather than at the deployed model [7, 42, 225, 270]. Next, *Interface-based AI-behavior control* gives the individual an interface through which to inspect, constrain, or modify the system’s behavior directly [51, 253, 267, 284]. Meanwhile, there are two types of human-centered steering. *Explanations, recommendations, and warnings* make the system’s decisions, reasoning, confidence, or limitations visible so that the person can decide what to do with them [17, 87, 212]. *Scaffolding human decisions or skills* uses the system as decision support that structures, supports, or in some cases impairs how the person decides, learns, or reflects [185, 209, 218, 262].

4.3.2 **Q2: Individual $\times$ Multi-Agent**. We report the codes for Q2 in Table 3 and elaborate on them below.

Represent. How do mechanisms represent an individual and the specialized agents that must jointly align to them? The Represent dimension spans four main codes. Some papers considered alignment to individuals by studying *Human representation and simulation* of human preferences, behaviors, and expectations within multi-agent systems [152, 171, 196], such as by learning human preferences on interaction data during training [152]. A different set of papers focuses on the representation of the agents in multi-agent systems. This includes *Agent-role and specialization representation* in multi-agent systems where agents are assigned specialized roles (e.g., planner, critic) [35, 215, 283] to improve alignment to human preferences [215, 283]. Other papers have a strong focus on *Cooperative orchestration*, which involves how multiple agents communicate and coordinate their work, including how they collaborate, supervise, or depend on one another within the interaction topology and its communication protocols [24, 160, 207]. The last set of papers focused on *Human-AI Relational Dynamics* by modeling the dynamics of human-AI interactions [76, 89, 136, 202]. The papers from human-computer interaction venues studied the effects of human interaction with multi-agent systems [76, 89], while machine learning venues modeled human interaction with multi-agent systems [136, 202].

Measure. How do mechanisms measure (mis)alignment between an individual and multiple interacting agents, and who judges? The Measure dimension is divided into five codes. Works measure alignment largely through *Task performance evaluations* involving understanding whether the multi-agent system

accomplishes a task through task outcomes or performance measures [85, 193, 207, 210], often using task-specific metrics (e.g., F1 score [244], success rate [193], human ratings [210]). Other codes involve using agents themselves to measure multi-agent alignment to individuals. This includes *Agent-based judging or verification*, when a work uses one or more agents to evaluate the behavior or outputs of other agents [54, 118, 175, 227]. Many of the works under this code study an agent’s effectiveness to be a judge [54, 175, 281]. A few other works examine *Multi-agent debate and critique* [35, 177] by studying the effectiveness of debate-based techniques for alignment. Finally, several papers analyze agent behaviors themselves. One code involves *Multi-agent interaction and coordination analysis* of communication patterns, interaction traces, and coordination processes among agents, such as agents’ perceptions of one another [239]. Another code studies *Collusion and failure-propagation detection* of undesirable or harmful collective behaviors among agents, such as lying [97, 145].

Steer. How do mechanisms orchestrate, constrain, or correct multiple agents on behalf of an individual, and steer the individual in return? The Steering dimension is split into eight codes. Most codes involve steering mechanisms by aligning agents to individuals. Several works investigate *Agent selection and task allocation*, which involves the selection of which agents to participate in a task [24, 48, 160]. *Multi-agent orchestration and communication design* examines how to design and control the ways in which agents interact while executing multi-agent tasks [35, 91, 131, 234], such as steering frameworks to guide agent behavior during multi-turn dialogues, disagreements, or debates [35, 91, 131]. Another common topic is *Adversarial steering and oversight*, which studied how to use adversarial agents, critic agents, or other forms of oversight to influence other agents through feedback or corrective interaction [28, 100, 216, 254]. Another code is *Agent-to-agent critique and correction*, which focuses on allowing agents to constrain or correct other agents through feedback [124, 137, 171, 205]. These works focus on more collaborative forms of steering techniques, such as teacher-student modeling [137, 151] or manager agents [171]. Next, the code of *Bounded agent autonomy* focuses on restricting agent decision spaces via constraints (e.g., budgets [256]) or permissions. *Simulation-based realignment* leverages individual simulations (e.g., LLM-generated synthetic data [214]) to generate adaptation signals to modify multi-agent systems. Next, *Human intervention and escalation interfaces* provides mechanisms for humans to intervene, override, or re-direct multi-agent decisions [231, 260, 263], such as model routing for human intervention [260]. Finally, only one code was related to aligning the human to multi-agent systems. *Human-facing information and cognitive steering* investigates how to steer human understanding, trust, or decision-making through explanations, evidence, transparency, feedback, or reflection [74, 130, 222, 256]. Many of these papers involve the design of human-agent interfaces [41, 222] or investigating the effects of agent behaviors on humans via user studies [130].

4.3.3 **Q3: Societal $\times$ Single-Agent**. We report the codes for Q3 in Table 4 and discuss them below.

Represent. How are diverse stakeholder values and knowledge represented to guide one focal AI system? The Represent dimension contains eight codes. *Structured Stakeholder Elicitation*

	Code	Definition	References
	<i>Whose state is represented, and in what form?</i>		
REPRESENT	Human representation and simulation	Representing human preferences, behaviors, expectations, etc.	[152, 171, 196]
	Agent-role and specialization representation	Defining specialized roles such as planner, critic, executor, or overseer / agent capabilities, expertise, limitations, and specialization within a multi-agent system.	[35, 215, 283]
	Cooperative orchestration	Representing how multi-agents communicate, collaborate, coordinate, supervise, or depend upon one another.	[24, 160, 207]
	Human-AI Relational dynamics	Modelling the dynamics of Human-AI interaction, partnership, interactions and collaboration.	[76, 89, 136, 202]
	<i>Against what criterion, on what evidence, judged by whom?</i>		
MEASURE	Task performance evaluation	Evaluating whether the focal AI system helps accomplish a task through task outcomes, success rates, accuracy, or other performance measures.	[85, 139, 193, 207, 210, 244]
	Multi-agent debate and critique	Evaluating multi-agent alignment through structured debate, critique, or mutual evaluation among agents.	[35, 177]
	Agent-based judging or verification	Using one or more agents to evaluate, verify, or audit the behavior, outputs, or decisions of other agents.	[54, 118, 175, 227, 258, 281]
	Multi-agent interaction and coordination analysis	Analyzing communication patterns, interaction traces, and coordination processes among agents.	[239]
	Collusion and failure-propagation detection	Detecting undesirable collective behaviors, including collusion, cascading failures, or harmful interaction patterns among agents.	[97, 145]
	<i>What is intervened on, through what channel, on whose authority?</i>		
STEER	Agent selection and task allocation	Selecting which agents participate in a task and routing subtasks based on agent capabilities, expertise, or current system states.	[24, 48, 160]
	Multi-agent orchestration and communication design	Designing and dynamically controlling how agents interact, including sequencing, activation, communication structures, and information-exchange rules during multi-agent operation.	[35, 85, 91, 131, 234]
	Adversarial steering and oversight	Using adversarial, critic, or oversight agents to influence other agents through feedback, constraints, or corrective interaction.	[28, 69, 100, 171, 216, 242, 254]
	Agent-to-agent critique and correction	Allowing agents to modify, constrain, or correct other agents through feedback and interaction.	[6, 124, 137, 151, 171, 200, 205]
	Bounded agent autonomy	Restricting agent decision spaces through constraints, permissions, or predefined operating boundaries.	[256]
	Human intervention and escalation interfaces	Providing mechanisms for humans to intervene, override, or redirect multi-agent decisions.	[231, 260, 263]
	Human-facing information and cognitive steering	Steering human understanding, judgment, trust, or decision-making through explanations, evidence presentation, transparency cues, feedback, or structured reflection.	[41, 74, 92, 130, 222, 256]
	Simulation-based realignment	Using individual simulations to generate adaptation signals that modify agent personas, policies, or behaviors toward desired societal outcomes.	[214]

Table 3: Level-3 codes of Q2 (Individual × Multi-Agent, 57 papers): mechanisms, codes, and definitions.

surfaces community needs through targeted methods, such as interfaces probing fairness [32] or case studies capturing LLM perspectives [282]. These insights are then translated into artifacts through *Participatory Stakeholder Co-Design*, for instance by empowering communities to collaboratively build data probes for advocacy [273]. *Community Knowledge and Taxonomy Construction* uses community-sourced knowledge bases to capture cultural nuances and prevent early classification decisions from inappropriately constraining downstream AI behavior [199, 224]. *Persona and Retrieved-Context Conditioning* leverages group interactions to match social norms [236] or utilizes demographic profiles to generate culturally aligned content [192]. *Persistent Norm and Objective Encoding* embeds values directly into system architecture, such as models grounded in regional cultural contexts [219]. In addition, *Deliberative Preference Construction* creates a collective preference or fairness standard through explicit stakeholder dialogue, such as voting with democratic deliberation to map opinion shifts [261]. At a broader societal scale, *Feedback- and Population-Based Preference Aggregation* aligns systems by weighing policies proportionally to true population distributions rather than majority opinions [104]. At the institutional scale, *Sociotechnical Governance Modeling* maps AI-stakeholder dependencies, such as through critical interventions for vulnerable populations [204] and collaborative participatory testbeds [271].

Measure. How is one focal AI system evaluated against diverse stakeholder expectations and its broader societal consequences? The Measure dimension contains seven codes. *Stakeholder-Grounded Participatory Evaluation* uses stakeholder needs, collaborative audits, and community feedback as evaluation evidence [38, 45, 86, 101, 183]. *Human Experience and Prospective Harm Evaluation* extends alignment measures longitudinally by examining changing perceptions, trust, and concerns for others' well-being over sustained interaction [96, 233]. On the model side, *Behavioral Robustness and Consistency Testing* examines how models respond to manipulation and counterfactual changes [141, 259], while *Group, Cultural, and Normative Behavior Comparison* examines behavioral differences across cultural contexts, populations, and moral standards [140, 167, 184]. Interpreting these evaluations requires interrogating the standards they encode. *Data and Benchmark Value-Coverage Auditing* examines whose perspectives and values evaluation resources reflect [154, 188], while *Mechanism, Intent, and Objective Auditing* examines how algorithmic objectives and system structures can produce misalignment [5, 59]. Beyond the individual, *Sociotechnical Dynamics and Impact Evaluation* examines the downstream distributions of social good and longer-term feedback involving beliefs and values, such as simulations and benchmarks that study social and behavioral changes [172, 173, 272].

Steer. How do interventions on stakeholder judgment, social interaction, and one focal AI system support alignment

	Code	Definition	References
REPRESENT	<i>Whose state is represented, and in what form?</i>		
	Structured Stakeholder Elicitation	Obtains explicit stakeholder values, needs, goals, or judgments through a predefined elicitation format.	[132, 138, 161, 182, 282]
	Participatory Stakeholder Co-Design	Uses collaborative design activity to turn situated stakeholder or expert knowledge into requirements, artifacts, or design proposals.	[12, 64, 157, 228, 273]
	Community Knowledge and Taxonomy Construction	Uses affected communities or domain experts to construct, revise, or audit categories, classifications, harm concepts, and representational priorities.	[99, 179, 199, 224, 278]
	Persona and Retrieved-Context Conditioning	Supplies demographic, viewpoint, interaction, or group context to the AI at prompting or retrieval time.	[135, 192, 236]
	Persistent Norm and Objective Encoding	Embeds cultural knowledge, ethical principles, or intertemporal preferences in persistent data resources, objectives, rewards, or lifecycle requirements.	[58, 168, 219]
	Deliberative Preference Construction	Produces a collective preference or fairness standard through explicit stakeholder dialogue, explanation, or negotiation.	[261]
	Feedback- and Population-Based Preference Aggregation	Produces a collective alignment signal from behavioral acceptance, iterative community feedback, or population-aware weighting.	[104, 142]
MEASURE	Sociotechnical Governance Modeling	Represents roles, dependencies, shared narratives, decision processes, or governance arrangements linking stakeholders, institutions, and the AI.	[146, 156, 189, 204, 271]
	<i>Against what criterion, on what evidence, judged by whom?</i>		
	Stakeholder-Grounded Participatory Evaluation	Uses stakeholder judgments, needs, or feedback to assess whether a focal system meets affected communities' expectations and priorities.	[38, 45, 86, 101, 183]
	Human Experience and Prospective Harm Evaluation	Assesses experienced or anticipated consequences of interaction with a focal system through people's perceptions, trust, control, well-being, or reported harms.	[96, 233]
	Behavioral Robustness and Consistency Testing	Tests whether model behavior remains safe, fair, or consistent under attacks, manipulation, counterfactual changes, or contextual pressures.	[18, 29, 83, 141, 221]
	Group, Cultural, and Normative Behavior Comparison	Compares model behavior across populations, cultural contexts, or specified normative standards to identify differences in alignment.	[140, 144, 167, 184, 220]
	Data and Benchmark Value-Coverage Auditing	Audits the inclusion, omission, or distortion of values and group perspectives in datasets and benchmarks, including whether their diversity is preserved in outputs.	[154, 188, 247, 251]
	Mechanism, Intent, and Objective Auditing	Examines objectives, prompting intent, oversight arrangements, or retrieval structures for design assumptions and failure modes that can produce socially consequential misalignment.	[5, 34, 59, 277]
STEER	Sociotechnical Dynamics and Impact Evaluation	Examines or anticipates downstream harms, distributional outcomes, and feedback between a focal AI system and societal actors, institutions, or evolving values.	[57, 66, 172, 173, 272]
	<i>What is intervened on, through what channel, on whose authority?</i>		
	AI-Mediated Judgment and Value Calibration	Uses AI interventions to help stakeholders clarify, compare, and reconcile judgments, values, preferences, or fairness criteria while humans retain or share the final normative choice.	[60, 181, 275]
	AI-Mediated Social and Strategic Influence	Uses AI interventions to shape stakeholder communication, participation, persuasion, norm-following, or other social responses in a group setting.	[61, 78, 186, 232, 237]
	Human-Specified Model Behavior Steering	Uses stakeholder values, rubrics, policies, viewpoints, incentives, or corrections to condition or update AI behavior so outputs better match specified targets.	[8, 122, 191, 195, 246]
	Sequential Feedback and Population Correction	Uses repeated human or population feedback to revise decisions, recourse, selection, or retraining over time so future outcomes better match a desired long-run objective.	[56, 170, 257, 268]

Table 4: Level-3 codes of Q3 (Societal $\times$ Single-Agent, 83 papers): mechanisms, codes, and definitions.

with societal values over time? The Steer dimension contains four codes. *AI-Mediated Judgment and Value Calibration* uses AI to help stakeholders clarify, compare, and reconcile evaluative standards while preserving human authority over the final normative choice, such as fair model selection by helping practitioners navigate competing fairness objectives [181]. *AI-Mediated Social and Strategic Influence* steers interaction among people, such as counterspeech bots that shape bystander responses to hateful discourse [232]. Meanwhile, other steer mechanisms act more directly on the focal model. *Human-Specified Model Behavior Steering* uses stakeholder-specified targets, policies, rubrics, personas, incentives, or collective-control principles to condition or update AI behavior. This includes reducing emergent allocation bias through explicit exploration incentives [246], updating models when policies or social norms change [191], and conditioning generation on data-driven personas [122]. Finally, *Sequential Feedback and Population Correction* contributes methods that intervene in society-AI feedback loops to keep long-term outcomes aligned with desired objectives.

This includes revising selection policies, reinforcement-learning policies, or retraining procedures in response to changes in human behavior or population dynamics [56, 170, 257, 268].

4.3.4 **Q4: Societal $\times$ Multi-Agent.** Below, we elaborate on its codes (see Table 5).

Represent. How do mechanisms represent human preferences, priorities, roles, and real-world institutional context? The Represent dimension has four codes. For representing human states, *Participatory and negotiated value representation* captures stakeholder values, preferences, lived experiences, and viewpoints through consultation, co-design, and other forms of stakeholder participation [49, 68, 147, 274]. *Sociotechnical relationship and topology modeling* captures relational states by modeling interaction structures, dependencies, and relationships among agents, institutions, and stakeholders [203]. Next, *Collective and institutional state representation* captures collective states, institutional conditions and governance contexts that shape alignment [148]. Finally, *Interdependent social-context modeling* captures the social, institutional

	Code	Definition	References
REPRESENT	<i>Whose state is represented, and in what form?</i>		
	<u>Participatory and negotiated value representation</u>	Representing stakeholder values and requirements through participatory elicitation, consultation, or co-design processes.	[49, 68, 147, 274]
	<u>Sociotechnical relationship and topology modeling</u>	Modeling interaction structures, dependencies, and relationships among agents, institutions, and stakeholders.	[203]
	<u>Collective and institutional state representation</u>	Representing collective states, institutional conditions, and governance contexts that shape multi-agent alignment.	[148]
	<u>Interdependent social-context modeling</u>	Modeling the social, institutional, and contextual conditions that shape interactions among multiple agents.	[44]
MEASURE	<i>Against what criterion, on what evidence, judged by whom?</i>		
	<u>Collective and social-impact outcome evaluation</u>	Evaluating how multi-agent interaction, social influence, coordination, and collective action affect group-level or societal outcomes.	[77, 108, 166, 201]
	<u>Distributional evaluation across populations and agents</u>	Evaluating how benefits, harms, responsibilities, and alignment outcomes are distributed across populations and agents within a multi-agent system.	[116, 165, 226]
	<u>Multi-agent interaction and coordination analysis</u>	Analyzing communication patterns, interaction traces, and coordination processes among agents.	[15, 75, 164]
	<u>Deliberative-quality evaluation</u>	Evaluating the quality of multi-agent deliberation, including consideration of competing perspectives, negotiation processes, and collective decision-making.	[169, 266]
	<u>Multi-agent institutional auditing and compliance</u>	Evaluating whether multi-agent systems collectively satisfy institutional requirements, governance standards, and accountability obligations.	[23, 46, 53, 84]
	<u>Emergent-risk and systemic-harm assessment</u>	Assessing emergent risks, cascading failures, and unintended consequences arising from interactions among multiple agents.	[190]
STEER	<u>Human feedback and outcome evaluation</u>	Evaluating alignment through feedback, judgments, and observed outcomes from humans or affected stakeholders interacting with a multi-agent system.	[98]
	<i>What is intervened on, through what channel, on whose authority?</i>		
	<u>Simulation-based realignment</u>	Using population-grounded or multi-stakeholder simulations to generate adaptation signals that modify agent personas, policies, or behaviors toward desired societal outcomes.	[4, 11, 27, 105, 238]
	<u>Incentive and mechanism design</u>	Designing incentive structures, allocation mechanisms, or institutional rules that shape how multiple agents behave, coordinate, and pursue collective objectives.	[112, 150, 174, 208, 265]
	<u>Multi-agent coordination protocols</u>	Establishing communication, interaction, and coordination rules that regulate how multiple agents exchange information and perform joint tasks.	[79, 93, 105, 119]
	<u>Distributed oversight and accountability</u>	Distributing monitoring, evaluation, and intervention responsibilities across multiple agents, institutions, or oversight entities.	[229, 276]
	<u>Institutional checkpoints and approval mechanisms</u>	Introducing institutionally defined checkpoints, approval requirements, or escalation rules that constrain when multi-agent actions can proceed.	[229]
	<u>Collective feedback integration</u>	Incorporating collective feedback, societal input, or aggregated preferences to modify multi-agent objectives, policies, or coordination strategies.	[279]
	<u>Institutional authority redistribution</u>	Changing the distribution of decision rights, influence, and control among agents, organizations, and stakeholders.	[25]
	<u>Negotiation and conflict-resolution protocols</u>	Designing procedures that enable agents with competing objectives or preferences to negotiate, resolve conflicts, and reach collective decisions.	[111]

Table 5: Level-3 codes of Q4 (Societal × Multi-Agent, 50 papers: mechanisms, codes, and definitions.

and contextual conditions that shape interactions among multiple agents [44].

Measure. How do mechanisms evaluate (mis)alignment among humans and agents across interactions, outcomes, and broader social standards? The Measure dimension has seven codes, typically relying on simulation outputs, interaction logs, and automated metrics. *Collective and social-impact outcome evaluation* asks whether coordination leads to equitable group outcomes [77, 108, 166, 201], while *Multi-agent interaction and coordination analysis* track how agents influence each other’s decisions across turns [15, 75, 164]. *Deliberative-quality evaluation* examines whether deliberation genuinely engages competing perspectives [169, 266]. Meanwhile, some research evaluates against external standards. *Distributional evaluation across populations and agents* checks whether LLM outputs reflect different cultural populations’ value priorities [116, 165, 226], whereas *Multi-agent institutional auditing and compliance* assesses compliance with governance or policy requirements [23, 46, 53, 84]. *Emergent-risk and systemic-harm assessment* targets systemic risks such as coordinated manipulation [190]. Finally, the primary human-centered mechanism

is *Human feedback and outcome evaluation*, measuring alignment through the judgments of the people actually affected [98].

Steer. How do mechanisms correct, adapt, or anticipate multitudes of human and agent interactions? The Steer dimension contains eight codes. Steering is conducted primarily through *Incentive and mechanism design* [112, 205], such as incentivizing truthfulness by semivalues and data valuation functions; *Simulation-based realignment* [4, 11], including symbolic judging agents that evaluate the ethics of multi-agent set-ups [27]; and *Multi-agent coordination protocols* [79, 119] through functional prototypes. Fewer cases use *Collective feedback integration* [279]. Meanwhile, other papers examine how alignment authority in these systems is divided through *Institutional authority redistribution* [25] while also identifying how authority is constrained through *Institutional checkpoints and approval mechanisms* such as gates, access, roles and approval requirements [93, 98, 276]. Across these mechanisms, work aims to increase collective bargaining power [25], consensus [111], and deliberated discussion [105] or to reduce large-scale effects such as polarization [128], model collapse [279], and unfair resource allocation [39].

5 Discussion

5.1 Using the BiALIGN PYRAMID Design Space

Scenario 1: Tutoring. Consider a team designing an AI tutor that adapts to one learner over a semester without creating dependence. Applying the BiALIGN PYRAMID top-down, the Universal Core identifies what the mechanism must do. The mechanism must *represent* the learner’s skills and reliance, *measure* how both change through use, and *steer* the interaction when reliance begins to undermine learning. Because both the learner and tutor may change through this process, its *direction* is mutual and the *timescale* is longitudinal. The setting then specifies the lower levels of the pyramid. One learner and one AI assistant place the scenario in Q1 (Individual × Single-Agent), where personalized state representation, experience-grounded evaluation, and direct behavioral attribution are relevant. The Q1 codes suggest using human state representation and context and memory construction to model the learner over time. Human behavioral evaluation and human–AI calibration assessments can measure independence alongside task success. Context-based adaptation and interface-based control can then steer the interaction.

This mapping exposes a weakness in a design that measures only whether the tutor produces correct answers. Task success would not reveal whether the learner is becoming more capable or simply more reliant on the tutor. Task success does not reveal whether the learner is becoming more capable or more reliant on the tutor. The BiALIGN PYRAMID pyramid, therefore, points to measures of the learner and the human–AI relationship, as well as steering mechanisms that act on the interaction rather than only the model. The design space can thus identify what an alignment mechanism is missing and guide its redesign, rather than only classify an existing system.

Scenario 2: Multi-agent healthcare system. A hospital system has several AI agents that are involved in a patient’s care. One agent supports triage and another recommends treatment, with earlier decisions shaping the information and options available later. The system affects patients and clinicians, but what counts as a good outcome may differ between them. At the Universal Core, alignment requires representing these interests and how decisions move through the system, then measuring whether the resulting care remains aligned with them over time. The *directional* changes are mutual over a longitudinal timescale. The involvement of multiple groups and multiple agents places the system in Q4 (Societal × Multi-Agent). Participatory or negotiated value representation provides a way to incorporate different stakeholder interests. Distributional evaluation can reveal when the system works differently for particular patient groups, while institutional auditing can examine how effects accumulate across agents. Steering may require institutional checkpoints that give people authority to intervene in decisions that cross agent boundaries.

Looking at each agent separately could miss the alignment problem entirely. The agents might each perform well by their own criteria while their combined decisions produce poor outcomes for some patients. Q4 shifts attention to what happens across the system and who can act when no single agent is responsible. Together, these examples show how the BiALIGN PYRAMID pyramid can be applied across different human and AI configurations to identify

relevant alignment mechanisms and reveal gaps in a system’s design. Section 5.2 examines how these mechanisms are distributed across the four quadrants in the literature.

5.2 Coverage and Gaps in the Literature

Coverage within each quadrant and patterns across them reveal where alignment mechanisms are concentrated and where gaps remain. Figure 5 shows the distribution of Represent, Measure, and Steer codes across these settings.

Within-quadrant trends. In Q1, representation focuses primarily on human and model states, while few systems maintain information that both the person and AI can access and update (12%). Measurement is relatively balanced across task performance, user experience, and model behavior. Steering more often changes the AI through fine-tuning or prompt adaptation than supports how people understand or work with AI through scaffolding, explanations, or interface controls. In Q2, representation shifts toward agent roles and relationships. Task performance and agent-based judging are the most common forms of measurement, while few papers examine collusion or failures that propagate across agents. Steering most often occurs through agents critiquing or correcting one another, although some systems provide information or controls to help people supervise them. Q3 more often involves people directly in defining what the system should represent, particularly through co-design and structured elicitation. However, few papers address how different preferences are combined or negotiated when people disagree. Measurement also focuses more often on model behavior than on people’s experiences or potential harms. Q3 differs from the other quadrants in steering: systems frequently seek to influence people through social or strategic information and support changes in their judgments or values. In Q4, representation most often involves people participating in or negotiating the values the system should reflect. Measurement focuses on collective outcomes and whether agents comply with institutional requirements, while fewer papers examine how outcomes differ across groups. Steering is dominated by incentive and mechanism design (44%), while institutional checkpoints, distributed oversight, and collective feedback each appear in only 4% of papers coded for steering.

Patterns across quadrants. Several patterns recur across the four settings. Simulated or constructed representations of people appear in every quadrant, but their role changes with the setting. They are used to judge model behavior in Q1, represent people interacting with multiple agents in Q2, and represent different people or groups in Q3. In Q4, simulation is used to test or adjust the behavior of multi-agent systems at the societal level. Its prominence is notable because decisions about how these systems should change can be based on simulated responses rather than direct input from affected people. Representations that people and AI can both access and change are also uncommon. They appear only in Q1, where they are the least common representation code. In the other quadrants, representations of people are primarily created for the system to use, while representations of agents support coordination. These patterns show that people are frequently represented in alignment mechanisms without having much control over those representations. Systems may rely on simulated or system-created

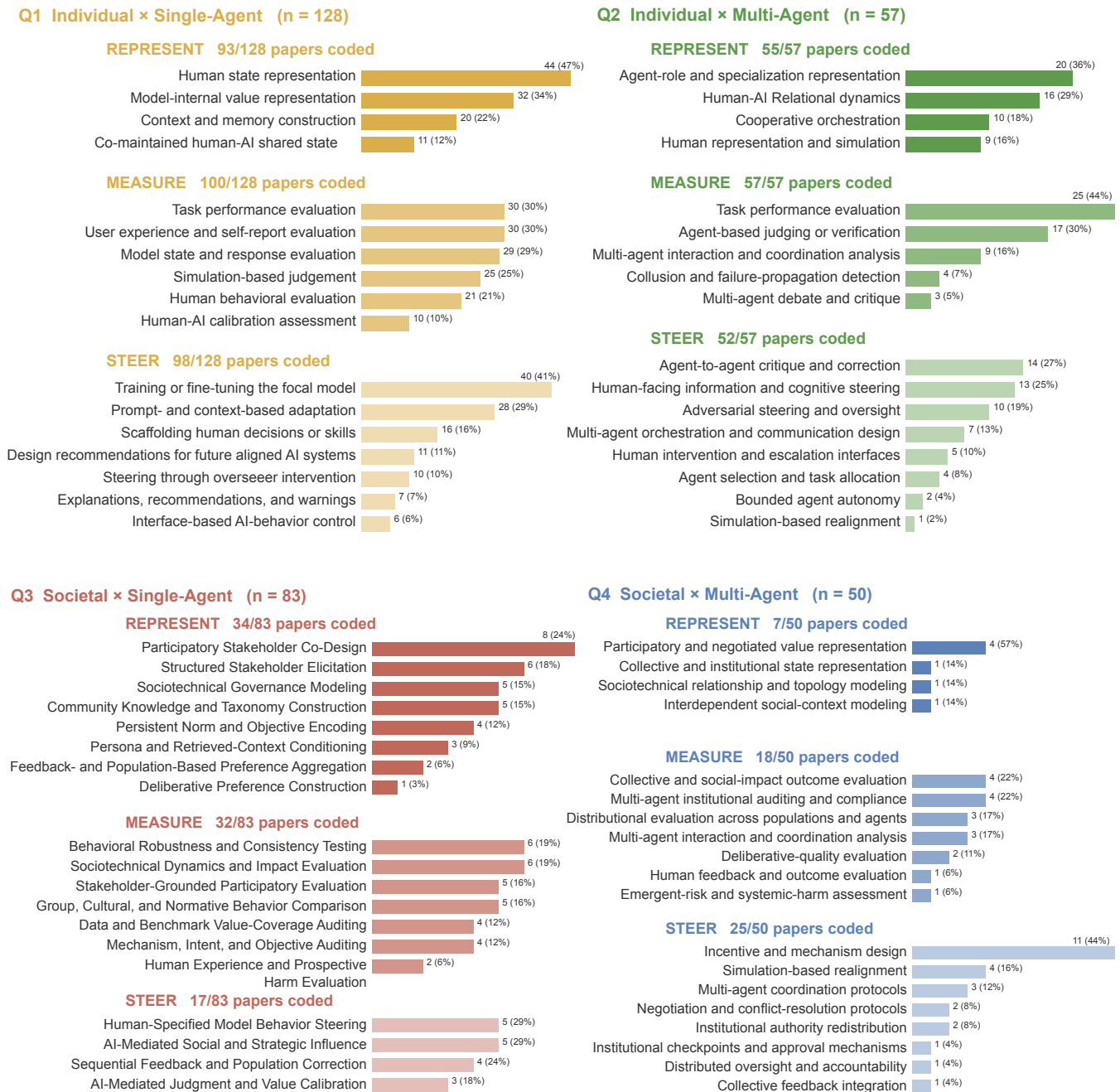


Figure 5: Distribution of Level-3 codes across the four quadrants. Each bar gives the number of papers carrying a code, grouped by stage. A paper may carry more than one code within a stage, so bars do not sum to the number of papers; percentages are taken over the papers coded at that stage, given beside each stage heading. The figure describes how the mechanisms in our corpus are distributed within the design space, not the prevalence of alignment research in general. Q1 and Q2 papers were coded on all three stages, whereas each Q3 and Q4 paper was assigned to a single stage; stage totals in the societal quadrants therefore reflect that assignment and are not comparable across quadrants.

representations of people, while giving people few opportunities to inspect or revise how they are represented as they change over time.

Directionality and timescale. The Level-3 codes in Figure 5 show which mechanisms are used to represent, measure, and steer alignment, while the paper-level facets in Figure 6 capture two properties that cut across those mechanisms: the direction in which change occurs and the timescale over which alignment is considered. Direction shows a similar pattern across all four quadrants (Fig. 6). Mechanisms designed to change the AI account for 72–81% of papers, while AI→Human mechanisms account for 15–23%. Mutual mechanisms, in which changes to people and AI can inform one another, are less common still and reach their highest share in Q4 (12%). For example, in Q1, steering is concentrated in fine-tuning and prompt adaptation, while scaffolding, explanations, and interface controls are less common. Q3 differs at the mechanism level, with more visible use of AI-mediated social influence and judgment or value calibration. Together, the two figures show that alignment is more often designed around changing the AI than mutual adaptation, while the mechanisms through which AI may shape people vary across settings.

Timescale changes more clearly across the quadrants. Among papers whose temporal horizon could be identified, session-level mechanisms are most common in the individual settings (45% in Q1 and 56% in Q2). Longitudinal mechanisms become more common moving toward societal and multi-agent settings, increasing from 28% in Q1 to 58% in Q4. Figure 6 establishes how long these mechanisms operate or are studied, but not whether observations made over time feed back into later interactions. A study may track changes in people or interactions without using those changes to update what the system represents or how later interactions are evaluated and steered. Longitudinal observation therefore does not necessarily create a feedback loop. Taken together, these findings show that alignment can be longitudinal without being bidirectional or dynamic. Studies increasingly account for change over time as systems become more distributed, but mutual adaptation remains rare, and observed changes do not necessarily shape what happens next.

These coverages and gaps reinforce the need to treat alignment as a property of the coupled human-AI system rather than the AI alone. Across the corpus, people are often represented but have limited ability to revise those representations, and alignment mechanisms more often adapt AI than support reciprocal change. The BiALIGN PYRAMID makes these gaps visible across different human and AI configurations and provides a basis for asking how alignment mechanisms need to change as those configurations scale.

5.3 Scaling Alignment Across the Axes

The quadrants illustrate how alignment changes as interactions scale from individuals to societies and from single agents to multi-agent systems. Each introduces a different alignment problem — interaction, delegation, pluralism, and governance—and changes what must be represented, measured, and steered. Below, we examine these shifts across the human and AI axes.

Individual × single agent as human-AI 1-on-1 interaction. This setting centers alignment on interaction: how do you and I align? This is the most studied quadrant in the corpus. The agent must understand what the person wants, while the person must understand enough about the agent to decide how and when to rely on it. Representation captures the person’s goals, preferences, or values and the context developed through interaction. Measurement considers both agent behavior and the person’s understanding of it. Task performance alone may not establish alignment if the person misjudges the agent’s capabilities or the agent does not reflect their intent, making calibration an important part of evaluation. Steering allows the person to correct divergence through instructions, interface controls, or changes to the model. In this one-to-one setting, the same person can provide the alignment target, evaluate the agent, and intervene when needed. This provides the baseline from which alignment changes as more people or agents enter the system. *While the human factors literature provides insights for this setting (for example, in the context of “trust calibration” or shared autonomy [153, 243]), there is a need to integrate multiple alignment targets together.*

Individual × multi-agent as task delegation. Moving from one to multiple agents introduces delegation: how does one person’s intent remain effective when work is distributed across agents they cannot inspect turn by turn? Outcomes now depend on how roles are allocated and how agents interact, rather than on a single decision locus. Representation must capture how the person’s intent applies across different agent roles. Measurement becomes less attributable to any one agent when failures emerge through interaction or propagate across agents [164, 166]. Agent judging and critique provide additional forms of oversight [35]. Steering expands from correcting an individual agent to changing how the crew operates, including task allocation, agent autonomy, and when control returns to the person. *The challenge is preserving human control when the person remains the source of intent but cannot supervise every action.*

Society × single agent as AI pluralism. Moving from an individual to society while holding the AI side to a single agent introduces another form of pluralism: how does one system reconcile what we want when there is no single account of what “we” want? A content moderation agent or an AI system supporting public-benefits decisions, for example, acts in relation to people who may disagree about appropriate or fair outcomes. Representation therefore shifts from eliciting one person’s preferences toward constructing a societal alignment target that accounts for conflicting values and unresolved differences. Deliberative processes can support collective positions while retaining differences among participants [261], and public input can be translated into principles that shape model behavior [81]. Measurement must consider whose perspectives are represented in the data and criteria used to define alignment [188]. Steering also changes because no single person can determine how the agent should behave. People need ways to challenge how their interests are represented and participate in revising the alignment target. *The challenge is constructing a collective alignment target from plural and sometimes conflicting values while preserving opportunities for disagreement and revision.*

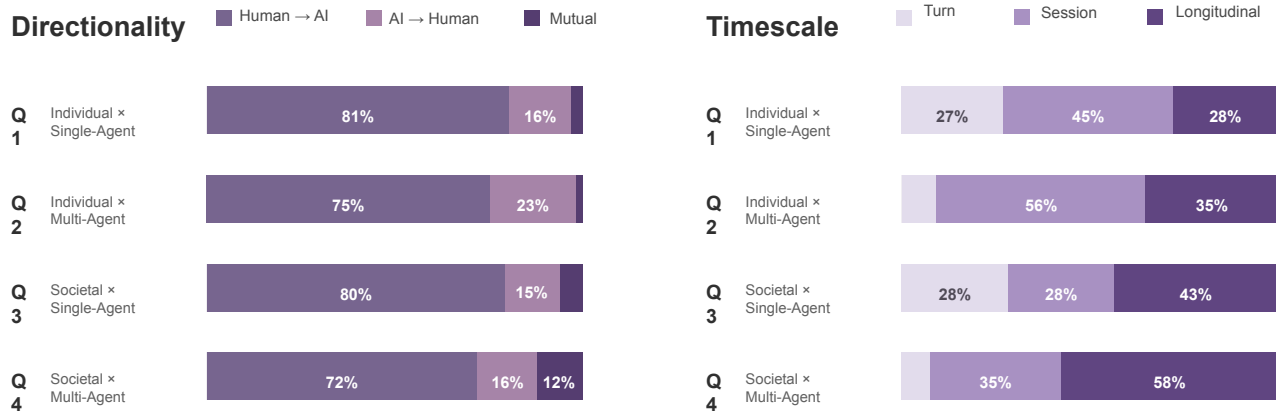


Figure 6: Paper-level facet coverage across the four quadrants. Left: Directionality — the mechanism’s designed endpoint of influence, as a share of each quadrant’s papers. Human→AI designs dominate every quadrant (72–81%), while *Mutual* designs remain rare: 3% in Q1, 2% in Q2, 5% in Q3, and 12% in Q4. **Right: Timescale** — the horizon over which a paper explicitly models change, computed over papers whose temporal horizon could be identified from the available evidence (papers labeled *Unclear* are excluded from the denominator). Turn-level shares are 27% in Q1, 9% in Q2, 28% in Q3, and 7% in Q4. Segments too small to label in the chart are reported here instead; percentages may not sum to exactly 100 due to rounding.

Society × multi-agent as AI governance. When both sides become distributed, the central problem shifts toward governance: how do we govern a socio-technical system in which both human interests and AI agency are distributed? AI governance provides structures and processes for determining how AI systems are developed and used responsibly [162]. In a healthcare system where several agents support decisions across patient care, for example, there is neither a single alignment target nor a single AI decision-maker. The pluralism problem now extends across both axes. Representation must account for where societal values enter the system and how they apply to agents with different roles. Measurement must connect outcomes to affected communities even when responsibility is distributed. Agreement among agents does not establish that those communities are represented. Steering raises questions of authority as well as control. The ability to intervene in an agent does not determine who should be allowed to do so, and changing one agent can affect behavior elsewhere in the system. Governance must establish who sets system goals, who can change particular agents, and who is responsible for effects that extend across them. Affected groups also need ways to challenge decisions and prompt revision. *The challenge is governing alignment when neither the interests being represented nor control over the AI system resides with a single actor.*

Taken together, these quadrants show why alignment must be understood at the level of the coupled human–AI system. As the number of humans or agents increases, direct interaction gives way to delegation, pluralism, and governance, and mechanisms that work in one setting may not transfer unchanged to another. The human and AI axes therefore help explain where Represent, Measure, and Steer mechanisms apply, what changes as the system scales, and why alignment cannot be reduced to the behavior of a single model.

5.4 Recurring Strategies and Where to Design Next

The four taxonomies in Tables 2–5 were developed independently, by separate coding teams, from disjoint sets of articles. The BiAlign Pyramid nonetheless surfaces similarities across quadrants: mechanisms that rest on the same underlying idea, and mechanisms that differ in implementation but overlap in their objectives. An overview of the four quadrants thus reveals recurring strategies and the areas they leave open for future work, discussed in this subsection.

Recurring mechanism types. Across the four quadrants, some representation mechanisms exhibit similarities. Each quadrant has a mechanism for representing human states through elicitation, inference from observable behaviors or traces, or simulation. In quadrants Q1 to Q3, values and objectives can be encoded into the AI system, using embeddings in Q1, agent roles in Q2, and persistent norms encoding in Q3. Measurement mechanisms show similar analogies. All quadrants evaluate observable behavior, of the model, the agents, or the humans involved, and identify potential sources of misalignment: task performance metrics (Q1, Q2) and behavioral robustness and consistency testing (Q3), collusion and failure-propagation detection (Q2) and emergent risk and systemic-harm assessment (Q4). Overseer entities are used across all quadrants, from crowd workers and LLM judges to stakeholder panels and institutional audits. Steering strategies show similarities as well. Every quadrant includes a channel for human override, from a person steering an agent’s decisions (Q1, Q2) to institutional, population or individual feedbacks and checkpoints (Q3, Q4). Longitudinal steering strategies include user edits, simulations, or population-level corrections. Interfaces in Q1, communication design in Q2, deliberation formats in Q3, negotiation protocols in Q4 are established as communication channels for steering.

Emerging differences. While the quadrants share some similarities across the dimensions, we highlight different strategies and techniques that differentiate them. First of all, individually and societally oriented systems derive their alignment objectives differently. Work in individual quadrants computes the objective from individual data traces, while work in societal quadrants convenes objectives through workshops, co-design and negotiation. Second, single agent and multi-agent systems rely on different measurement techniques. Numerical alignment metrics such as the task success rates and accuracy scores serve as evidence of alignment in Q1 and Q2. These mechanisms have no counterpart at the societal level, as Q3 and Q4 assess systems through audits and stakeholder judgments. Conversely, audits are rare for individual-centered systems, where personalization and training pipelines are more frequently evaluated through performance indicators. Lastly, steering strategies differ in where model updating sits. Explicit parameter updating is a distinct, dominant code only in Q1 (training or fine-tuning, 41%); in the other quadrants it appears within broader codes — sequential feedback and population correction in Q3, simulation-based realignment in Q2 and Q4 — as one option among interaction-, incentive-, and protocol-level interventions. Fine-tuning stratified for societal-level alignment goals (e.g., [279]) therefore remains an underexplored design.

Current gaps. Based on the overview of the design space, we highlight four main areas for future work. First, we highlight the under-representation of longitudinal designs across the whole design space. Longitudinal, self-improving systems introduce risks especially for societal-level, complex system designs.

Because explicit model updating dominates only in Q1, long-term alignment in the other quadrants depends on multi-level steering such as inspectable artifacts and institutional gates that prevent the recursive loop from degrading into sycophancy or collective manipulation. Second, we problematize the potential over-reliance on simulation as a tool to represent individual users, evaluators, and overseer entities, as well as societal-level actors. This is especially evident in Q4, where societal actors are largely addressed through simulated agents rather than real populations. Since simulated stakeholders express objectives their designers gave them, this form of representation might reinforce unchecked assumptions. Third, we highlight an under-represented emphasis on observability as a tool for bidirectional alignment. The current corpus shows that training data and methods are often unavailable, and measurement and steering mechanisms leveraging complex computation or AI models remain challenging to interpret. On the one hand, explainable steering mechanisms like Q1’s *explanations, recommendations, and warnings* could reduce reliance on opaque AI systems. On the other hand, mechanisms that surface how the humans are affected by AI systems could increase user awareness. Lastly, we note that a bidirectional communication channel like the Q1 mechanism *co-maintained human-AI shared state* is unique among in the quadrants. Shared artifacts between humans and AI could become significant anchors of alignment goals during long-scale tasks, in part due to their visibility to both humans and AI systems.

5.5 Limitations and Future Work

Our corpus is focused rather than exhaustive. It covers eleven venues from 2021 through the available 2026 proceedings, while the interaction-and-adaptation gate excludes some static techniques and work from other disciplines or emerging practice. Passing this gate does not establish bidirectionality or loop closure; Direction and Timescale describe separate properties. Moreover, coverage-preserving sampling in the individual–single-agent quadrant retained conceptual diversity rather than the original frequency distribution. Corpus counts therefore indicate descriptive coverage, not unbiased prevalence across alignment research. The individual–societal distinction also compresses intermediate scales such as teams, organizations, classrooms, and communities, while real systems may span or transition between quadrants. Finally, the Pyramid organizes mechanisms without prescribing which combinations are appropriate. Its usefulness and normative implications—including consent, contestability, and legitimate authority — depend on stakeholders and institutional contexts and should be evaluated through participatory and longitudinal applications in future work.

6 Conclusion

We introduced the BiALIGN PYRAMID, a multilevel design space for organizing human–AI alignment mechanisms across human scales and AI configurations. The design space combines three complementary alignment functions—Represent, Measure, and Steer—with four cross-setting bands, four configuration-specific quadrants, and the cross-cutting properties of Direction and Timescale. Mapping a focused corpus of 318 papers shows that alignment is operationalized differently as systems move from individual interaction to societal governance and from single models to multi-agent configurations. It also identifies limited coverage of reciprocal adaptation, operational feedback across repeated interactions, and representations that affected people can inspect and revise. These patterns do not imply that every alignment mechanism is bidirectional or dynamically closed. Rather, they show why direction, temporal horizon, and loop closure should be treated as explicit design choices. The pyramid provides a shared vocabulary for comparing these choices and for identifying opportunities for future human–AI interaction research.

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A Author Roles and Contributions

This work is the product of a large, distributed collaboration. The project was organized into four coding teams, each coordinated by team leaders, with project advisors providing guidance on framing and the manuscript, and a project leader and principal investigator overseeing the overall effort. Within each role group, authors are listed alphabetically by last name. Table 6 summarizes each author's self-reported participation in six project activities, and Section A.1 gives individual contribution statements.

A.1 Individual Contribution Statements

Four Teams' Leaders.

Giulia Frascaria co-led Team 1 with Daniel Hiterer and helped Team 2 with their coding. She coded approximately 40 papers across all dimensions, was a main developer of the Team 1 codebook (with Daniel Hiterer and Shengchen Zhang), proposed ideas for the use case and visualization in the paper, and contributed to writing Sections 4 and 5.

Daniel Hiterer co-led Team 1 with Giulia Frascaria and was a main developer of the Team 1 codebook. He contributed to primary and IRR coding, proposing ideas in team meetings, writing, paper ideation, and reviewing.

Tiffany Kneareem co-led Team 2. She annotated papers and performed IRR coding, proposed ideas in team meetings, wrote portions of the paper, and reviewed and polished the draft prior to submission.

Jenny T. Liang co-led Team 2. She contributed to primary and IRR coding, proposing ideas in team meetings, writing portions of the draft, paper ideation, and reviewing.

Kaifang Mao helped Team 2 annotate 10 papers, completed the paper filtering stage, organized the Level 3 codes, and contributed to writing Section 5.1, in addition to primary and IRR coding, team discussions, and paper ideation.

Yike Shi co-led Team 3. She contributed to all project activities: primary and IRR coding, proposing ideas in team meetings, writing portions of the draft, paper ideation, and reviewing and polishing the paper.

Dasha Simons co-led Team 4 with Anoop K. Sinha, alternating in running team meetings, structuring the work, establishing the team's ways of working, and developing the paper's ideas and approach. She contributed to the coding and to writing Team 4's section of the paper, provided feedback on other teams' drafts, and attended and contributed to most team-leader meetings.

Anoop K. Sinha co-led Team 4 with Dasha Simons. He contributed to IRR coding, proposing ideas in team meetings, writing portions of the draft, paper ideation, and reviewing.

Hongwei Zhu co-led Team 3. He annotated 17 papers as primary coder and 15 as secondary coder, participated in team-leader meetings, and contributed to team discussions, writing, and reviewing.

Team Members.

Salvatore Andolina contributed primarily to the coding and actively participated in essentially all Team 1 meetings and discussions. He also maintained and improved the shared

coding spreadsheet (e.g., fixing formulas, reorganizing the sheet, and adding columns) and contributed to the final review and polishing of the paper.

Jessica Y. Bo participated in all rounds of coding for Team 3, took part in team meetings, and reviewed the writing.

Giulia Di Fede coded 32 papers across all three dimensions, participated in reconciling codes on Miro, re-coded papers as primary and secondary coder during the formalization of the codebook, and resolved disagreements in the final phases of the kappa calculations.

Helena Anna Frijns contributed to primary and IRR coding, proposing ideas in team meetings, writing, and reviewing.

Eric Lacosse performed multiple rounds of coding on assigned papers and participated in both sub-team and all-hands meetings, raising concerns and suggesting ideas to facilitate discussion of what bidirectional alignment should be.

Sachit Mahajan annotated 11 papers and contributed to team discussions on coding decisions, including refining low-level mechanism labels, consolidating higher-level codes, resolving ambiguous Represent/Measure/Steer classifications, and harmonizing the team's codes with the final cross-team framework. He also helped identify and describe exemplar papers and contributed to writing Section 4.3 and the Discussion.

Srishti Palani contributed to primary and IRR coding, proposing ideas in team meetings, paper ideation, and reviewing and polishing the paper.

Nikolas Papadopoulos contributed to primary and IRR coding (18 papers), proposing ideas in team meetings, writing and reviewing assigned sections for Team 4 (Multi-Agent $\times$ Society).

Weiyang Shi helped Team 4 conduct the first-round literature review by identifying and reviewing relevant papers for the Multi-Agent $\times$ Society topic, contributed to the definition of the MEASURE category and the time-scale taxonomy (single-turn, multi-turn, and longitudinal), annotated 11 papers, completing both the main coding and the IRR coding and drafted assigned sections of the paper.

Sajal Surati annotated 40 papers as primary coder and 40 papers as secondary coder, and proposed ideas in team meetings.

Nadav Viduchinsky led the analysis for the MEASURE section of the Society $\times$ Single-Agent group: identifying initial themes and codes, reviewing the aggregated codes, re-analyzing the data with the aggregated codebook, and carrying out the final coding pass to establish inter-rater reliability. He also wrote that section and populated its data tables.

Konstantinos Voudouris contributed to the Team 2 annotations (primary and IRR coding) and to reviewing the paper.

Katherine Wang contributed to primary and IRR coding, proposing ideas in team meetings, writing, paper ideation, and reviewing.

Eryue Xu coded 10 papers as primary annotator and 5 as secondary annotator, actively discussed codes with members working on the same area (Steer $\times$ Session), contributed points to the discussion, and reviewed and edited the drafts.

Table 6: Self-reported author contributions by activity. • = completed; ◦ = in progress or committed at the time of reporting; blank = not involved. *Primary*: team-based paper coding as primary coder; *IRR*: team-based inter-rater reliability coding as secondary coder; *Ideas*: actively proposing ideas in team meetings; *Writing*: writing portions of the draft; *Ideation*: paper framing and brainstorming of writing and analysis ideas; *Review*: paper reviewing and polishing.

Author	Primary	IRR	Ideas	Writing	Ideation	Review
<i>Team Leaders</i>						
Giulia Frascaria	•	•	•	◦	◦	◦
Daniel Hiterer	•	•	•	◦	◦	◦
Tiffany Kneareem	•	•	◦	◦	◦	◦
Jenny T. Liang	•	•	•	•	•	◦
Kaifang Mao	•	•	•	◦	•	◦
Yike Shi	•	•	•	•	•	•
Dasha Simons	•	•	•	◦	◦	◦
Anoop K. Sinha		•	•	•	◦	◦
Hongwei Zhu	•	•	◦	◦		◦
<i>Team Members</i>						
Salvatore Andolina	•	•	◦	◦		◦
Jessica Y. Bo	•	•	•			◦
Giulia Di Fede	•	•				◦
Helena Anna Frijns	•	•	◦	◦		◦
Eric Lacosse	•	•	•		◦	
Sachit Mahajan	•	•	•	◦	◦	◦
Srishti Palani	•	•	•		•	•
Nikolas Papadopoulos	•	•	•	•		◦
Weiyang Shi	•	•	•	•	•	•
Sajel Surati	•	•	•			
Nadav Viduchinsky	•	•	•	•		◦
Konstantinos Voudouris	•	•				•
Katherine Wang	•	•	◦	◦	◦	◦
Eryue Xu	•	•	•	◦	◦	◦
Bhada Yun	•	•	•	◦	◦	◦
Shengchen Zhang	•	•	•		•	◦
<i>Project Advisors</i>						
Yang Li			◦		◦	◦
Tanu Mitra			•	◦	•	◦
Brad A. Myers					◦	◦
Thiemo Wambsganss			◦	◦	◦	◦
Philipp Wintersberger			•	◦	◦	•
<i>Project Leader and Project Investigator</i>						
Hua Shen	•		•	•	•	•

Bhada Yun represented the STEER sub-team for Team 4 (Multi-Agent × Society), helped refine the terminology and the codebook, and annotated 9 papers.

Shengchen Zhang annotated 23 papers as primary annotator and 30 as secondary annotator; contributed to discussions about the workflow, codebook construction, and code definitions, as well as organizational tasks early in the collaboration; suggested “steer” as a more neutral framing than “intervention” for the design space dimensions; and helped review and polish the paper.

Project Advisors.

Yang Li contributed ideas in project meetings, to the paper’s ideation and framing, and to reviewing the paper.

Tanu Mitra contributed ideas in project meetings and to the paper’s ideation and framing, and contributed to writing and reviewing.

Brad A. Myers was part of the original workshop organizing team, contributed to the original design space through the workshop and the organizers’ initial meetings, recruited the leadership for Team 1, and contributed to paper ideation and reviewing.

Thiemo Wambsganss served in a supervisory role, contributing feedback, ideas, writing, and reviewing; he also coded 4 papers at the beginning of the project.

Philipp Wintersberger scheduled team work in the early phase of the project, discussed and reviewed the approaches to coding and the early codebook, and contributed to reviewing, editing, and drafting the final manuscript.

Project Leader and Project Investigator (Project PI).

Hua Shen led the project as project leader and principal investigator. She contributed to primary paper coding, proposing ideas in team meetings, paper ideation and framing, writing the manuscript, and reviewing and polishing the paper.

B Systematic Literature Review

B.1 Systematic Literature Review Process

To identify research on dynamic and bidirectional human-AI alignment, we conducted a systematic literature review following the PRISMA guidelines and established recommendations for literature reviews in HCI [159, 213]. As shown in Figure 2, the review involved paper identification, multi-stage screening, corpus construction, and qualitative coding. We describe each stage below.

B.1.1 Identification and Screening with Keywords. We started with papers published from January 2021 through the latest available 2026 proceedings across eleven major venues in HCI, NLP, machine learning, and responsible AI (see Appendix B.2). We retrieved 89,241 papers in the initial identification stage. We then screened their titles and abstracts using a predefined list of keywords related to dynamic and bidirectional human-AI alignment (see Appendix B.3). We retained 36,561 papers containing at least one keyword or its variants for further screening.

B.1.2 Assessing Eligibility with Four Gates. We further filtered the 36,561 papers using four gates. These gates assessed whether a paper: 1) studied a setting in which humans and AI systems interact with or influence one another; 2) contributed a mechanism for representing, measuring, or steering human-AI alignment; 3) examined how human-AI relationships evolve over time; and 4) considered bidirectional influence or feedback between humans and AI systems. Papers passing the first three gates formed an intermediate corpus of 6,232 papers. Applying the fourth gate resulted in a core corpus of 2,128 papers on dynamic and bidirectional human-AI alignment. We applied the four gates at scale using LLM-assisted screening, with human verification for ambiguous and conceptually important cases. Appendix B.4 provides the gate definitions; the prompts and validation procedure are reported in Appendix C.

B.1.3 Assessing Centrality and Constructing the Final Corpus. We organized the 2,128 core papers along two dimensions: the human level (individual or societal) and the AI configuration (single- or multi-agent). This produced four groups: individual-single-agent, individual-multi-agent, societal-single-agent, and societal-multi-agent. We then assessed each paper’s centrality to dynamic and bidirectional human-AI alignment as less relevant, relevant, or highly relevant. This assessment identified 867 highly relevant papers: 677 individual-single-agent, 57 individual-multi-agent, 83 societal-single-agent, and 50 societal-multi-agent papers.

Because the individual-single-agent group was substantially larger than the other groups, we applied coverage-preserving sampling to this group while retaining all papers in the other three groups. We organized the 677 individual-single-agent papers into 126 fine-grained categories across Represent, Measure, and Steer, and used a greedy set-cover procedure to select up to three representative papers per category. This reduced the group from 677 to

Table 7: Inter-rater reliability for Level-3 coding. Entries are macro-average Cohen’s κ values across non-constant binary codes. Paired papers are reported as Represent/Measure/Steer; the stage mean equally weights the three stage-level coefficients.

Quadrant	Paired papers	Macro-average Cohen’s κ			Stage mean
		Represent	Measure	Steer	
Q1	91 / 87 / 92	0.724	0.725	0.806	0.752
Q2	54 / 57 / 52	0.948	0.984	0.847	0.926
Q3	32 / 32 / 16	0.864	0.896	0.798	0.853
Q4	7 / 18 / 24	0.692	1.000	0.906	0.866

128 papers, resulting in a final corpus of 318 papers for qualitative coding.

B.1.4 Qualitative Code Development. We first conducted qualitative coding for each paper by identifying relevant passages that addressed the research question associated with Represent, Measure, or Steer, and recording Level 1–3 codes in a shared codebook. Each annotator was assigned to one of the three stages and received a Principal list of papers for primary annotation. Annotators reviewed the relevant sections of each paper and independently generated codes without viewing the existing AI-generated labels, thereby ensuring that the initial code development reflected their own interpretation of the literature.

After completing the independent annotation, annotators compared their codes with the AI-generated Level 3 candidate labels and resolved discrepancies by adding a relevant label they had missed, merging semantically equivalent labels, or rejecting an unsupported AI label. We then aggregated the annotators’ codebooks and consolidated overlapping and related concepts to derive the final Level 3 categories. Using the finalized codebook, we re-coded all papers at Level 3; Levels 1 and 2 were not re-coded. Each annotator subsequently applied the Level 3 codebook to their Supporter list, providing the papers assigned for reliability assessment with a second independent annotation. We assessed agreement between the two annotations using inter-rater reliability. The finalized Level 3 annotations were used to map the reviewed literature onto our design space and to support the analyses presented in the main paper. Detailed agreement results are provided in Appendix B.1.5.

B.1.5 Inter-Rater Reliability for Level-3 Coding. For each non-exclusive Level-3 code, we computed Cohen’s κ as a binary present/absent decision and macro-averaged the coefficients over non-constant codes within each stage and quadrant. Table 7 reports the resulting stage-level reliability estimates. The paired-paper denominator varies by stage because reliability was assessed on the papers assigned to that stage.

B.2 Venues

We focused on archival research papers published between 2021 and the latest available 2026 proceedings across HCI, NLP, machine learning, and responsible AI. We included main-track long and short papers, Findings, PACMHCI journal-track papers, the NeurIPS Datasets and Benchmarks track, and NLP industry-track papers.

We excluded workshops, late-breaking work, extended abstracts, posters, demos, tutorials, doctoral consortium papers, competitions, and other non-archival tracks. The venues included in our review were:

- **HCI:** CHI, CSCW, and UIST;
- **NLP:** ACL, EMNLP, and NAACL;
- **Machine Learning:** ICLR, NeurIPS, and ICML;
- **Responsible AI:** FAccT and AIES.

We retrieved structured metadata from the ACL Anthology for NLP venues and from OpenReview for machine learning venues. For CHI, UIST, and FAccT, we obtained paper records from DBLP and supplemented missing abstracts using OpenAlex, Semantic Scholar, and Crossref. We retrieved CSCW papers from PACMHCI through OpenAlex and AIES papers through OJS OAI-PMH supplemented by OpenAlex. We merged duplicate records using DOI and normalized-title matching before keyword screening.

B.3 Keywords

We developed a high-recall keyword set covering the human and AI entities involved in alignment, the phenomena through which alignment may be expressed, and the mechanisms used to represent, measure, explain, or steer alignment. We organized the keywords into two scope filters (Human and AI), six topic anchors (Alignment, Failure/Dynamics, Value, Interact, Safety, and Ethics), and four phase categories (Representation, Evaluate/Measure, Explain, and Steer). We matched keywords against paper titles and abstracts using case-insensitive whole-word or whole-phrase matching, including morphological variants where appropriate. Within each category, a paper matched if it contained at least one keyword. A paper passed the keyword screening if it matched both scope filters and at least one topic or phase category: $\text{HUMAN} \wedge \text{AI} \wedge (\text{TOPIC ANCHOR} \vee \text{PHASE})$. The complete keyword set is listed below.

Scope filters.

Human. **Core:** human, user, person, people, individual. **Hyponyms:** patient, clinician, physician, doctor, nurse, surgeon, therapist, caregiver, investor, trader, broker, borrower, lender, shareholder, banker, taxpayer, policyholder, pedagogy, pedagogical, classroom, instructor, educator, tutoring. **Scale:** community, group, collective, society, societal, population, public, crowd. **Aspect:** cognition, cognitive, behavior, behavioral, perception, mental model. **Associated:** human-in-the-loop, human-centered, stakeholder, participant, participatory.

AI. **Core:** AI, artificial intelligence, machine learning, ML, model, algorithm. **Model type:** large language model, LLM, GPT, generative, foundation model, neural network, deep learning, NLP, VLM, multimodal, in-context learning. **Agent:** agent, agentic, multi-agent, MAS, autonomous, LLM agent. **Technique:** reinforcement learning, RL, chatbot, assistant, copilot, conversational AI. **Associated:** system, automation, automated, computational.

Topic anchors.

Alignment. **Core:** align, alignment, aligned, misalignment, misaligned. **Methods:** RLHF, RLAIF, DPO, reward model, reward modeling, preference learning, preference optimization, instruction tuning, instruction following, constitutional AI. **Direction/type:** bidirectional, bidirection, dynamic alignment, human feedback, human preference. **Associated:** value alignment, preference alignment, goal alignment, intent, intention.

Failure/Dynamics. **Failure modes:** sycophancy, sycophantic, manipulation, manipulate, overreliance, over-reliance, value drift, delusion, delusional. **Dynamics:** co-evolution, coevolution, co-adaptation, coadaptation, dynamic, dynamics, drift, evolution, temporal, longitudinal, long-term, trajectory-level. **Mechanism:** feedback loop, emergent, emergence, reward hacking, specification gaming, Goodhart, deception, deceptive. **Associated:** echo chamber, filter bubble, polarization, homogenization, model collapse.

Value. **Core:** value, values, principle, constitution, norm, norms, moral, morality, belief, beliefs. **Type/theory:** human values, social norm, ethical principle, Schwartz, value system, pluralistic, pluralism, value pluralism. **Elicitation:** preference, preferences, opinion, attitude, intent, intention. **Associated:** cultural, culture, ideology, political.

Interact. **Core:** interact, interaction, interactive, interacting, collaboration, collaborate, collaborative, conversation, conversational, dialogue, communication. **Type/role:** human-AI, human-computer, human-machine, human-agent, assistant, copilot, partner, partnership, teaming. **Temporal:** sustained, repeated, ongoing, continual, continuous. **Associated:** engagement, adoption, usage, interface, user experience, UX.

Safety. **Core:** safe, safety, unsafe, AI safety, safety-critical. **Risk/harm:** risk, harm, harmful, harmfulness, hazard, threat, vulnerability, misuse, abuse, malicious. **Safeguard:** robust, robustness, guardrail, oversight, scalable oversight, red-teaming, jailbreak, adversarial, corrigibility. **Associated:** reliability, reliable, security.

Ethics. **Core:** trust, trustworthy, trustworthiness, bias, biased, fair, fairness, ethical, ethics, responsible. **Dimensions:** accountability, transparency, discrimination, equity, justice, privacy, autonomy. **Societal/governance:** societal impact, social impact, governance, regulation, policy. **Associated:** misinformation, disinformation, persuasion, influence, social good.

Phases.

Representation. **Core:** represent, representation, model (as a verb), encode, capture. **Component:** user model, user modeling, preference modeling, value representation, persona, profile, preference elicitation, elicit, elicitation. **Associated:** infer, inference, estimate, formalize, specification, specify.

Evaluate/Measure. **Core:** measure, measurement, evaluate, evaluation, assess, assessment. **Component:** monitor, monitoring, track, tracking, detect, detection, metric, benchmark, audit, probe, quantify. **Associated:** analyze, measure drift, diagnostic, test, testing.

Explain. Core: interpretability, interpretable, explain, explanation, explainability, XAI. **Component:** transparent, transparency, understand, understanding, sensemaking, mental model, rationale. **Associated:** insight, clarify, reveal, communicate.

Steer. Core: steer, steering, steerability, intervene, intervention, guide, guidance. **Component:** control, controllable, correct, correction, adapt, adaptation, personalize, personalization, customize, customization, nudge. **Associated:** modify, adjust, influence, shape, feedback.

B.4 Screening Gates

Following keyword screening, we assessed each paper using four gates that progressively narrowed the corpus from broad human–AI research to work on dynamic and bidirectional human–AI alignment.

G1: Human–AI Coupling. A paper passed G1 if it studied humans interacting with an AI system or contributed a mechanism that used human-derived values, preferences, or norms as an alignment target. Papers with no human reference, as well as human-subject studies involving no AI system, failed this gate.

G2: Phase Mechanism. A paper passed G2 if it contributed a technique, system, method, framework, or measurement instrument for at least one alignment phase: Represent, Measure, or Steer. Papers that only observed, described, or discussed a phenomenon without contributing a mechanism failed this gate.

G3: Trajectory Relevance. A paper passed G3 if it examined a human–AI relationship at any timescale, including a single interaction, a session, a longitudinal process, or population-level change. Context-free capability benchmarks that measured no human–AI relational property failed this gate.

G4: Bidirectionality. A paper passed G4 if its mechanism addressed at least one of three forms of bidirectionality: AI-to-human influence, a feedback loop in which human responses shaped subsequent AI behavior, or change over time in the human, the AI, or their relationship. Mechanisms that treated human information solely as a fixed, offline input failed this gate and were retained as background alignment literature.

C LLM Screening Prompts and Human Validation

C.1 LLM-Assisted Screening Procedure

We used Claude Opus 5 to apply the four screening gates. For G1–G3, the model received each paper’s title, venue, publication year, and abstract and returned decisions for the three gates, an overall verdict, the teaming classification, alignment phase, timescale, and a one-sentence rationale. G4 was subsequently applied to papers passing G1–G3 and returned a bidirectionality decision, direction label, corpus tier, and rationale. We instructed the model to prioritize recall when the available information was ambiguous.

C.2 Screening Prompts

C.2.1 Prompt for G1–G3. The model was given our working definition of alignment: “A human–AI system is aligned when its co-evolutionary trajectory preserves humans’ reflectively endorsed values and agency, at both the individual and collective level.” It was then instructed to apply the following gates and include a paper only if all three passed:

G1: Coupling. The paper must address human–AI coupling through either a human party interacting with an AI system or a mechanism that represents or measures human values, preferences, norms, or their enactment by AI. A human-derived normative standard must be an input to the mechanism, or the resulting representation must be intended to stand in for humans in the coupled system. Human values appearing only as motivation or framing do not satisfy this gate.

G2: Phase contribution. The paper must contribute a technique, system, method, framework, or measurement instrument for Represent, Measure, or Steer. Papers that merely observe, describe, or discuss a phenomenon without contributing a mechanism do not satisfy this gate.

G3: Trajectory relevance. The paper must examine the human–AI relationship at any timescale, including a single turn, session, longitudinal process, or population-level change. Per-response alignment properties count as single-turn trajectory mechanisms. Context-free capability benchmarks that measure no human–AI relational property do not satisfy this gate.

The model returned binary decisions for G1–G3; an include, exclude, or maybe verdict; the teaming classification; the alignment phase and timescale; and a one-sentence rationale. A paper was included only when all three gates passed. The model was instructed to use maybe only when the abstract was genuinely insufficient and otherwise to favor inclusion when uncertain.

C.2.2 Prompt for G4. For papers passing G1–G3, the model was instructed to determine whether the paper’s mechanism engaged the reverse or coupled direction of alignment through at least one of the following:

- (1) **AI-to-human influence:** measuring or intervening on how AI shapes human beliefs, values, behavior, reliance, skills, well-being, or agency;
- (2) **Feedback loop:** using human responses to update the AI’s subsequent behavior, including interactive correction, iterative preference elicitation, co-adaptation, or personalization based on ongoing use; or
- (3) **Trajectory dynamics:** modeling, measuring, or steering change over time in the human, the AI, or their relationship, including value or preference drift, norm shifts, longitudinal adaptation, alignment decay, or societal diffusion.

The operational test asked whether the human side was treated only as a static input, such as a fixed dataset, benchmark, or one-shot preference corpus, with no human response or temporal change represented in the mechanism. Such papers failed G4 and were retained as background alignment literature. The model returned the G4 decision, the bidirectionality direction, the core or background tier, and a one-sentence rationale.

C.3 Human Validation Setup

We conducted human validation on 100 papers sampled from the corpus retained by the LLM-assisted screening. Two annotators independently assessed G1–G4, the teaming classification, and centrality. Because the validation sample was drawn from papers that passed the automated screening, pass decisions were highly prevalent and the marginal label distributions were imbalanced. For G1–G4, we therefore report observed agreement and Gwet’s AC1, a chance-corrected coefficient designed for settings with high agreement and shown to be less sensitive than Cohen’s κ to prevalence and marginal-distribution effects [67, 245]. For teaming classification, we report observed agreement and Cohen’s κ . For centrality, whose labels were also imbalanced, we report observed agreement and Gwet’s AC1.

C.4 Human Validation Results

For the four screening gates, human–human agreement ranged from 0.80 to 0.88, with Gwet’s AC1 ranging from 0.766 to 0.871. These coefficients fall within the substantial (0.61–0.80) to almost-perfect (0.81–1.00) ranges commonly used to interpret chance-corrected agreement [114]. Human–AI agreement ranged from 0.77 to 0.96, with Gwet’s AC1 ranging from 0.743 to 0.959 across the four gates, indicating substantial to almost-perfect consistency between the LLM screening decisions and human judgments. Teaming classification likewise showed substantial human–human agreement ($\kappa = 0.722$) and substantial to almost-perfect human–AI agreement ($\kappa = 0.720$ and 0.813). For centrality, Gwet’s AC1 ranged from 0.696 to 0.783 across human–human and human–AI comparisons, indicating substantial agreement despite imbalanced label distributions [67, 245]. Together, these results support both the consistency of the human screening criteria and the reliability of the LLM-assisted classifications.

Table 8: Human–human and human–AI validation results ($N = 100$). Reliability is reported using Gwet’s AC1 for the screening gates and centrality, and Cohen’s κ for teaming classification.

Assessment	Item	Human 1–Human 2		Human 1–AI		Human 2–AI	
		Agreement	Reliability	Agreement	Reliability	Agreement	Reliability
Screening gates	G1	0.80	AC1 = 0.766	0.90	AC1 = 0.890	0.77	AC1 = 0.743
	G2	0.87	AC1 = 0.861	0.91	AC1 = 0.902	0.96	AC1 = 0.959
	G3	0.88	AC1 = 0.871	0.92	AC1 = 0.917	0.93	AC1 = 0.928
	G4	0.82	AC1 = 0.793	0.83	AC1 = 0.815	0.88	AC1 = 0.873
Teaming classification	Overall	0.80	$\kappa = 0.722$	0.86	$\kappa = 0.813$	0.79	$\kappa = 0.720$
Centrality	Overall	0.76	AC1 = 0.696	0.82	AC1 = 0.783	0.80	AC1 = 0.754